

Does Ageist Language in Job Ads Predict Age Discrimination in Hiring?

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We study ageist stereotypes reflected in job-ad language and age discrimination in hiring, exploiting job-ad text and evidence on age discrimination from a correspondence study. We develop and use methods from computational linguistics and machine learning. We find that language related to stereotypes of older workers sometimes predicts hiring discrimination against older men. This is the case for all three categories of age stereotypes we consider—health, personality, and skill. For women, we find that age stereotypes about personality predict differential hiring by age. The evidence for men is quite consistent with the industrial psychology literature on age stereotypes.

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I. Introduction

We develop and implement methods to explore the role of stereotypes in hiring discrimination using the text of job ads and apply these methods to evidence on age discrimination. We make two contributions. First, we develop techniques that leverage machine learning and textual analysis to analyze the text data in job ads from a large-scale field experiment on discrimination. Second, we use these techniques to produce evidence on which age-related stereotypes appearing in job ads are associated with an experimental measure of hiring discrimination against older workers—the first evidence we know of that establishes relationships between age-related stereotypes and actual employer behavior. This analysis provides evidence on whether employers who use ageist language in their job ads also have less intent to hire older workers—as captured in our experimental results.

The hypothesis underlying this analysis is that employers might use stereotyped language in job ads to shape the applicant pool, discouraging older workers from applying and reducing the likelihood that age discrimination is detected. Alternatively, different jobs may have different requirements, which could be stated in job ads, and employers may hold stereotypes about older job applicants' abilities to meet these job requirements—for example, assuming that older workers are less able to do heavy lifting. In either case, we might expect that employers who use job-ad language expressing ageist stereotypes are more likely to engage in age discrimination in hiring, with the latter case representing pure statistical discrimination.¹

Age discrimination is of great policy interest to the United States and other countries. Rapidly aging populations coupled with lower labor force participation of older individuals imply rising dependency ratios and strains on the finances of public programs targeted at older individuals, especially retirement and health care programs. As a result, there is an imperative to increase the employment of older individuals. The hiring of older individuals is likely an essential part of the solution. Nearly half of older workers move to “bridge” jobs or “partial retirement” jobs (Johnson, Kawachi, and Lewis 2009) before transitioning to complete retirement, and many leave retirement to take jobs before retiring again (Maestas 2010). Age discrimination may hinder the ability of older individuals to move into new jobs or reenter the workforce.

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¹ We address the possibility that if employers influence the applicant pool strongly enough, they may not have to hire at lower rates from the older applicants who do apply.

The most credible evidence on discrimination in hiring comes from field experiments—more specifically, resume-correspondence studies (Fix and Struyk 1993; Bertrand and Duflo 2017; Gaddis 2018; Neumark 2018). These studies have been applied to discrimination based on race, ethnicity, gender, age, and other group membership (e.g., disability). Resume-correspondence studies of age discrimination create fictitious but realistic job applicants who are on average equivalent except for age, which is signaled through school graduation year(s). Researchers use these fictitious job applicants to apply for real job openings, and age discrimination in hiring is measured by comparing interview request rates (“callbacks”) between older and younger applicants. Previous resume-correspondence studies almost always point to substantial age discrimination in hiring.² Recently, we conducted a large-scale field experiment studying age discrimination in hiring, focusing on potential sources of bias in past studies. We found compelling evidence of age discrimination—especially against older women (Neumark, Burn, and Button 2019).³

Our goal in the present paper is to advance the experimental literature on age discrimination in a direction that helps us understand what underlies age discrimination, delving inside the black box of why or how employers discriminate on the basis of age. Specifically, we use the text of job ads from Neumark, Burn, and Button (2019) to explore whether—and if so which—age stereotypes are associated with actual discrimination by employers. This inquiry is motivated by research in industrial psychology (and related areas), discussed in detail below, documenting that employers and others have negative stereotypes about older workers—such as lower ability to learn, less adaptability, worse interpersonal skills, less physical ability, lower productivity, worse technological skills and knowledge, and less creativity—all of which can deter their hiring. However, little is known about which stereotypes employers act on, if any, when making actual hiring decisions. The industrial psychology literature primarily uses surveys given to small samples of students or a general population about their attitudes concerning older individuals, but not necessarily in employment contexts, let alone the specific context of older workers seeking new jobs. Although some studies survey managers with hiring experience, in their actual roles as managers they may not use these stereotypes in making decisions. In addition, survey respondents may not honestly reveal discriminatory preferences,

² See Bendick, Jackson, and Romero (1997), Bendick, Brown, and Wall (1999), Riach and Rich (2006, 2010), Lahey (2008), Baert et al. (2016), Farber, Silverman, and von Wachter (2017), Farber et al. (2019), Carlsson and Eriksson (2019), Neumark, Burn, and Button (2016, 2019), and Neumark et al. (2019).

³ Neumark, Burn, and Button (2019) provide an extensive discussion regarding the interpretation of resume-correspondence study findings as reflecting age discrimination. Here, we simply interpret the evidence this way and refer readers to that paper for discussion of this issue.

stereotypes, or values if they are socially undesirable (e.g., Barnett 1998; Krumpal 2013).

For these reasons, in this paper we pursue evidence on the importance of age-related stereotypes for actual labor market behavior. We provide, to our knowledge, the first study that links age stereotypes to evidence on actual age discrimination in hiring.⁴ We use the text from more than 11,000 job advertisements from our field experiment, and we explore what job-ad language related to age stereotypes predicts age discrimination in hiring—as measured in the experiment by the younger applicant being called back but not the older applicant. For example, one stereotype of older workers is that they are not as good with technology (McCann and Keaton 2013). Job ads could contain language related to this stereotype (e.g., “must be a technological native”). We can then ask whether job ads containing such language are less likely to result in callbacks for older job applicants.

We find evidence that language related to stereotypes of older workers often predicts discrimination against older men.⁵ For men, we find evidence that age stereotypes about all three categories we consider—health, personality, and skill—predict age discrimination, and for women, age stereotypes about personality predict some differences in hiring by age. In general, the evidence is much stronger for men, and our results for men are quite consistent with the industrial psychology literature on age stereotypes.⁶

The machine learning and textual analysis methods we develop can be used to leverage text data in audit and correspondence studies of discrimination in labor markets and in other markets.⁷ These methods build significantly

⁴ There are other studies that find a link, albeit less directly, between age discrimination in hiring and age stereotypes. Carlsson and Eriksson (2019) conduct a resume-correspondence study and ask employers about stereotypes, finding that employers in their survey think that older workers have lower ability to learn new tasks, are less flexible/adaptable, and have less ambition. But they do not directly link the hiring outcomes to these survey responses about stereotypes. Van Borm, Burn, and Baert (2021) survey recruiters about their beliefs regarding the skills and abilities of hypothetical candidates in a vignette study. They find that employers view older workers as having worse skills and that the perceived differences in skills explained more of the discrimination against older women than older men that is suggested by the vignette study.

⁵ Of course, our methods do not speak to the role of stereotypes held by employers that are not reflected in job ads.

⁶ As discussed later in the paper, there is some evidence from industrial psychology and related research of stereotypes that are favorable to older workers and some that stereotypes that can either favor or disfavor them. We discuss the evidence on these stereotypes as well.

⁷ The only other studies that we are aware of that used textual analysis, with machine learning, on job ads are Jaeger et al. (2020) and Tian and Zhang (2021). Jaeger et al. apply machine learning to the text from ads to classify internships into occupation categories. Tian and Zhang apply machine learning to text from ads for

on earlier research leveraging text data from job ads, applying machine learning methods to the text analysis rather than just searching for key phrases.⁸ Our approach allows researchers to analyze text data when the phrasing is complex, varied, and not always obvious (e.g., the numerous ways one could phrase “communication skills”) and can also be applied to a wider range of empirical research questions in economics. We constructed our approach to create an *a priori* classification of the language that can be developed independently of the analysis of the relationship between the coding of language and the outcomes of interest. Researchers doing future correspondence studies or other types of studies who wish to utilize the text of the ads or other sources of information could preregister the application of and “output” from this method before collecting the data.

Our evidence on whether employers who use ageist language in their ads are less likely to hire older workers has implications for policy responses to reduce age discrimination. For example, job training, job coaching, or educational campaigns can focus on addressing the relevant negative stereotypes.

library science positions in China to detect explicit discriminatory preferences, including age limits for applicants. Some correspondence studies use textual data on a more limited basis. Hanson, Hawley, and Taylor (2011) study subtle discrimination through “keywords” used by landlords responding to prospective tenants. Hanson et al. (2016) had research assistants code the helpfulness and other characteristics of mortgage loan originator responses to prospective borrowers. Tilcsik (2011) identifies four words in job ads related to masculine stereotypes (“decisive,” “aggressive,” “assertive,” and “ambitious”) and links them to hiring outcomes in a study of discrimination against gay men. Nunley et al. (2015) uses textual analysis of job titles in ads to identify jobs requiring extensive customer interaction. Correspondence studies in other markets could also make use of text data. For example, Button et al. (2020) study discrimination in access to appointments with mental health professionals, and Ameri et al. (2020) study discrimination in access to Airbnb rentals. Both studies use platforms in which there is text data that could potentially be analyzed.

⁸ There are several notable examples of researchers using textual data in job ads outside of a resume-correspondence study. Kuhn and Shen (2013) and Kuhn, Shen, and Zhang (2020) explore how gender preferences feature explicitly or implicitly in job ads in China, and Hellester, Kuhn, and Shen (2020) explore age and gender preferences in job ads in China and Mexico. Modestino, Shoag, and Balance (2016) use text data from job ads to document that during the recovery from the Great Recession, “downskilling” occurred, with firms reducing skill requirements in job ads. Deming and Kahn (2018) use text data in job ads to measure how 10 different skills relate to wages. Marinescu and Wolthoff (2020) match text data from job ads to job application data to study the matching process between jobs and applicants. Banfi and Villena-Roldán (2019) use unique features of a job board to study how posted wages affect job applicants. Ash, Chen, and Ornaghi (2020) measure gender stereotyped language in judges’ decisions on gender-related issues (e.g., workplace outcomes, sexual harassment). In an early small study, Wax (1948) found that summer resorts in Ontario, Canada, were more likely to discriminate against Jewish customers (based on names) requesting accommodations if they used phrases like “restrictive clientele” in their advertising.

Efforts could also focus on improving hiring practices, perhaps by increasing the information available to employers that reduces the attribution of stereotypes to older workers. More generally, the Code of Federal Regulations covering the Age Discrimination in Employment Act (ADEA) currently states, “Help wanted notices or advertisements may not contain terms and phrases that limit or deter the employment of older individuals. Notices or advertisements that contain terms such as age 25 to 35, young, college student, recent college graduate, boy, girl, or others of a similar nature violate the Act unless one of the statutory exceptions applies” (§1625.4).⁹ Thus, our work can provide information to agencies that enforce age discrimination laws on job-ad language that may predict employer discrimination in hiring.

II. A Model of Stereotyped Language in Job Ads and Employer Discrimination

We develop a stylized model to help clarify how to interpret evidence that stereotyped language in job ads is or is not related to age discrimination. This is useful because it is conceivable that older workers’ responses to stereotyped job ads could end up obscuring the relationship between stereotyped language and measured discrimination. We are not studying applicants’ responses to stereotyped job-ad language because we do not observe these responses.¹⁰ But we consider the implications of this potential response for our empirical analysis.

Let h^T ($0 \leq h^T < 1$) index age discrimination in hiring by discriminatory employers. The parameter h^T is the relative hiring rate from the pool of potential old applicants compared with potential young applicants. The T superscript indicates that h^T is the “target” ratio of old versus young employees. Employers achieve their target employment ratio by applying the relative hiring rate h to the actual applicant pool.¹¹ Nondiscriminating employers have $h^T = 1$, so they would hire in proportion to the ratio of old versus young among potential applicants. We assume that nondiscriminating

⁹ An extreme version of such language is stating maximum experience levels in job ads—as occurred recently in *Kleber v. Carefusion Corp.*—which will clearly act to exclude many older applicants (see http://www.aarp.org/content/dam/aarp/aarp_foundation/litigation/pdf-beg-02-01-2016/kleber-amended-complaint.pdf; viewed November 8, 2017). Button (2020) discusses the ruling in this case.

¹⁰ There is research suggesting that, in other contexts, job seekers respond to job-ad language, including Belot, Kircher, and Muller (2018) and Banfi and Villena-Roldán (2019) on posted wages; Kuhn, Shen, and Zhang (2020) on gender requests; and Ibanez and Riener (2018), Leibbrandt and List (2018), and Flory et al. (2021) on affirmative action or diversity statements in recruitment materials or job ads.

¹¹ For purposes of discussion, we equate the callback rate with the hire rate. These may differ in a fixed proportion, so that $h = \theta \cdot cb$ (where h = old/young hire rate, cb = old/young callback rate, and $0 < \theta < 1$). The substance of this is that there is no manipulation of callbacks to young vs. old applicants to achieve hiring goals.

employers write age-neutral ads. In contrast, discriminatory employers may write stereotyped ads that might discourage older workers from applying.

If a discriminatory employer runs an age-neutral ad, we assume that the old/young ratio among actual applicants equals the ratio among potential applicants. In this case, the employer has to apply a lower relative hiring rate ($b = b^T$) to the older applicants. Alternatively, the employer can run stereotyped ads. This lowers the relative share of the old among actual versus potential applicants, allowing employers to hit their target ratio with hiring rate b closer to 1 than is b^T . If the response to the stereotyped ads is strong enough, the share of the old among actual applicants could be driven so low that employers do not need to use $b < 1$ to hit their target ratio. They could conceivably even favor older applicants ($b > 1$) if the response to stereotypes is large enough.¹² Thus, given the potential response to stereotypes, there may be a bias against finding discrimination in hiring rates for employers running stereotyped ads; that is, our estimates provide a lower bound. Conversely, however, we cannot get spurious evidence in favor of this hypothesis.

We would expect more potential applicant responses to stereotyped ads the higher the cost of applying is. In the (lower) limit, if applying is costless, applicants ignore the stereotyped ad language. Thus, there is less bias the lower the cost of applying. (And we think the cost of applying on our job board is quite low.) We would also expect more potential applicant response when the probability of being hired declines more in response to age-stereotyped language. This has implications for the potential bias in our estimates.

To develop these ideas more explicitly, we cast them in the context of our regression model. With nondiscriminatory hiring and no shaping of the applicant pool, the ratio of old hires (H_O) to young hires (H_Y) equals the ratio of old potential applicants (PA_O) to young potential applicants (PA_Y), because the nondiscriminatory hiring rate ($r = r_Y = r_O$) is applied to both older and younger potential applicants, implying

$$\frac{H_O}{H_Y} = \frac{r \cdot PA_O}{r \cdot PA_Y} = \frac{PA_O}{PA_Y}. \quad (1)$$

Discriminatory firms are defined by the parameter b^T , such that $H_O/H_Y = b^T \cdot PA_O/PA_Y$. There are two tools to achieve this goal: discriminatory hiring and using stereotyped language in the job ad. If a discriminatory firm engages in discriminatory hiring but the job-ad language does not deter

¹² A reviewer raises the possibility of a more complex response that changes the skill composition of older vs. younger applicants. For example, job ads may contain language about communications skills, which older workers perceive as a stereotype, causing those with worse communications skills to apply at a lower rate. It is beyond the ability of our data to test for variation in effects based on skills of applicants that are potentially related to age stereotypes, but this is an interesting question for further research.

older applicants, then $r_O/r_Y = b^T$, $H_O = r_O \cdot PA_O = r_Y \cdot b^T \cdot PA_O$, $H_Y = r_Y \cdot PA_Y$, and $H_O/H_Y = b^T \cdot PA_O/PA_Y$.

Stereotyped language in a job ad reduces the actual older applicants (AA_O) below PA_O . For example, suppose the effect of the stereotyped language is to drive AA_O to equal $b^T \cdot PA_O$. In that case, the firm can apply the nondiscriminatory hiring rate r to both groups, and again $H_O/H_Y = (r \cdot AA_O)/(r \cdot PA_Y) = b^T \cdot PA_O/PA_Y$.

We estimate models that, at their simplest, are of the form

$$d_{\text{exp}} = \alpha + \beta S + \varepsilon. \quad (2)$$

The parameter S is a measure of the “age stereotype” strength of the ad.¹³ In our implementation, S is a continuous variable that runs from -1 to 1 , but for what follows, we treat it as rescaled so that $S \geq 0$ (without loss of generality). The term d_{exp} is the experimental indicator of discrimination—for most of our analysis, whether the young applicant received a callback and the old one did not. This can be viewed as a job-ad-level measure of H_O/H_Y .

The question is whether a positive estimate of β indicates that firms that use ageist stereotypes in their job ads are more likely to discriminate. The potential problem is that if stereotyped job ads reduce AA_O relative to PA_O , age discrimination in hiring may not be accurately reflected in the estimate of β . When high- S job ads reduce applications from the old, AA_O/PA_O declines. This has a negative effect on d_{exp} because discriminating firms that receive fewer applications from older workers engage in less discrimination against older workers who apply. Thus, the estimate of β is biased downward—against finding evidence that firms that use age-stereotyped ads discriminate against older workers.

This bias is stronger the larger the response of applicants (AA_O/PA_O) to S is. We would think that this application response is higher the stronger the relationship between S and hiring discrimination against older workers is. This is helpful because it implies that the bias from the response of older applicants if anything obscures evidence that employers who use age-stereotyped ads discriminate against older workers.

To see this, suppose the utility of job j to person i is $U_{ij} = \varepsilon_{ij}$, where $\varepsilon_{ij} \sim N(0, 1)$.¹⁴ The cost of applying for a job is c . The probability of getting a job offer (callback) is $p_y = b$ ($0 < b \leq 1$) if young and $p_o = b(S)$ ($0 \leq b(S) \leq b$, $b'(S) < 0$) if old. That is, the probability of a job offer is lower the more age stereotyped the job ad is. An example of a function with $0 \leq b(S) < b$ is

¹³ It can be a vector corresponding to many stereotypes, but here we treat it as a single variable.

¹⁴ The variance can be fixed without loss of generality.

$$b(S) = b \cdot e^{-\eta S} \quad (\eta > 0). \quad (3)$$

A young person applies if $b \cdot \varepsilon > c$ (dropping the i and j subscripts) or $\varepsilon < -c/b$. Given the distributional assumption, the probability of applying is $A_y = \Phi(-c/b)$. An old person applies if $b(S) \cdot \varepsilon > c$ (dropping the i and j subscripts) or $\varepsilon < -c/b(S)$, so $A_o = \Phi(-c/b(S))$.¹⁵

For old applicants,

$$\frac{\partial A_o}{\partial S} = \frac{-(-c)b'(S)}{b(S)^2} \cdot \phi\left(\frac{-c}{b(S)}\right), \quad (4)$$

where ϕ denotes the standard normal density. Since $b'(S) < 0$, $\partial A_o / \partial S < 0$, establishing that there will be a negative response of older applicants to more stereotyped job ads. We would expect $|\partial A_o / \partial S|$ to be larger (a larger reduction in A_o) the more the probability of an offer declines with S . To see this, assume the specific functional form $b(S) = b \cdot e^{-\eta S}$, so that $b(S) = b$ if $S = 0$, and declines toward zero as S increases.

In this case, since $b'(S) = -\eta \cdot b \cdot e^{-\eta S}$,

$$\frac{\partial A_o}{\partial S} = \frac{(-c)\eta}{b} \cdot e^{\eta S} \cdot \phi\left(\frac{-c}{b} \cdot e^{\eta S}\right) \quad (5)$$

and

$$\begin{aligned} \frac{\partial^2 A_o}{\partial S \partial \eta} = & \left\{ \left(-\frac{c}{b}\right) \cdot e^{\eta S} - \frac{c}{b} \cdot \eta \cdot S \cdot e^{\eta S} \right\} \phi\left(-\frac{c}{b} \cdot e^{\eta S}\right) \\ & - \frac{c}{b} \cdot \eta \cdot e^{\eta S} \cdot \phi'\left(-\frac{c}{b} \cdot e^{\eta S}\right). \end{aligned} \quad (6)$$

Since for the standard normal $\phi'(x) = -x \cdot \phi(x)$,

$$\frac{\partial^2 A_o}{\partial S \partial \eta} = \phi\left(-\frac{c}{b} \cdot e^{\eta S}\right) \cdot \left\{ -\frac{c}{b} \cdot e^{\eta S} - \frac{c}{b} \cdot \eta \cdot S \cdot e^{\eta S} - \left(\frac{c}{b}\right)^2 \cdot \eta \cdot (e^{\eta S})^2 \right\}, \quad (7)$$

which is negative because $b, c, \eta, S > 0$.

Thinking back to our regression model (eq. [2]), β and η are positively related. When η is larger, the true relationship between age-stereotyped ads and discrimination against older applicants is stronger. The implication of $\partial^2 A_o / \partial S \partial \eta < 0$, then, is that there will be stronger negative bias in the estimate of β when the true value of β (and hence η) is larger.

This is significant because it implies that bias from the response of older applicants if anything obscures evidence that employers who use age-stereotyped

¹⁵ We assume that $\partial A_y / \partial S = 0$, or young people do not respond to the stereotyped language. We could have the probability of an offer for a young applicant increasing in S —i.e., the opposite direction from old people—and the qualitative conclusions are the same.

ads discriminate. This occurs because when the relationship captured by β (relating measured age discrimination to stereotyped ads) is weaker, the bias from the applicant response is also weaker. This makes sense because the bias comes from potential applicants responding to the likelihood that the stereotyped ad implies it is harder for an older applicant to get hired.

A failure to find evidence linking job-ad language to employer discrimination could reflect something else: (i) methods that are uninformative at detecting ageist stereotypes in job ads, (ii) employers not believing that job seekers respond to age stereotypes in job ads, or (iii) discriminatory employers simply do not use stereotyped language in job ads to discourage older workers from applying. In none of these cases is there an actual link between ageist stereotypes in job ads and employer age discrimination in hiring. Even more important, because there is a bias against finding evidence of such a link, evidence against the null would imply that ageist language in job ads does help predict hiring discrimination against older workers.

III. Background and Data from the Resume-Correspondence Study

Our experimental measure of discrimination comes from Neumark, Burn, and Button's (2019) large and comprehensive resume-correspondence study of age discrimination. The study used triplets of realistic but fictitious resumes for young (aged 29–31), middle-aged (aged 49–51), and older (aged 64–66) job applicants, sending 40,223 applications (resumes) to 13,371 job positions in 12 US cities (in 11 states). This is by far the most extensive resume-correspondence study of hiring discrimination to date, and the large number of job ads included in the study is critical to the methods we use in the present paper.

Neumark, Burn, and Button (2019) sent applications for positions in occupations that, based on Current Population Survey data, older as well as younger individuals often take as new jobs (hence likely bridge jobs for older workers): administrative assistant and retail sales jobs for women and retail sales, security, and janitor jobs for men.¹⁶

¹⁶ Neumark, Burn, and Button (2019) sent three applications per position: always one younger applicant and two older applicants of different ages (49–51 or 64–66) or with different work experience histories. While some of the resumes sent were on average identical to isolate the effect of age, as in the usual resume-correspondence design, Neumark, Burn, and Button (2019) also sent some older worker resumes with longer, more realistic work histories; arguably these applicants are more comparable to the younger applicants because their experience is commensurate with their age—like for young applicants. This was done to avoid the possibility of bias toward finding evidence of age discrimination, as older workers would not normally have the same listed work experience as younger workers. Neumark, Burn, and Button (2019) also used different resume types to explore whether older workers who exhibit “bridging” behavior—moving from demanding jobs or jobs with more responsibility

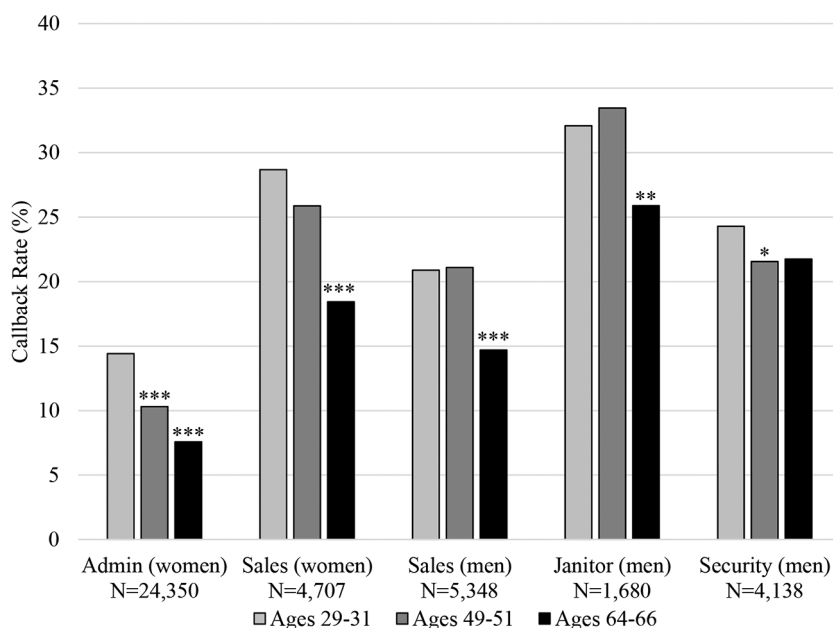


FIG. 1.—Comparisons of job applicant callback rates by age. A callback is defined as an interview invitation or similar positive interest in an applicant. The figure is reproduced from Neumark, Burn, and Button (2017) using data from Neumark, Burn, and Button (2019). One, two, and three asterisks indicate that the estimate is statistically significant at the 10%, 5%, and 1% level, respectively.

Figure 1 presents the main descriptive evidence from Neumark, Burn, and Button (2019). Across all occupations and genders, older applicants (age 64–66) got fewer callbacks than younger applicants. (These differences were statistically significant in all cases, except for men applying for security jobs.) As figure 1 shows, the magnitude of the discrimination against older women was larger. Neumark, Burn, and Button (2019) present several more sophisticated analyses, but the basic conclusion remains the same.

IV. What Can Our Evidence Tell Us about Discrimination?

Why might employers use stereotyped language in job ads, and what might this predict for our analysis? One hypothesis is that employers who discriminate on the basis of age use stereotyped language to shape the applicant pool, reducing the likelihood that age discrimination is detected. In particular, language that conveys positive stereotypes about young workers might discourage older workers from applying (as might language conveying

to jobs that are more flexible or with less responsibility—experienced more discrimination. Measured discrimination was generally insensitive to these variations.

negative stereotypes about older workers—although negative stereotypes are less common in our data). This would lead to the underrepresentation of older applicants in the applicant pool.

Why is this valuable to a discriminating employer? Presumably, the probability of an age discrimination claim (and an adverse outcome for the employer) depends on how much lower the ratio of job offers to applicants is for older applicants than for younger applicants (b , from the model). Then for the same number of older and younger hires, an employer who uses stereotypes that discourage older job applicants would have a lower probability of facing a discrimination claim.¹⁷ Thus, we can test the hypothesis that discriminatory employers use ageist language in job ads by relating the measure of age discrimination in the resume-correspondence study (differences in callbacks for older vs. young applicants) to age stereotypes in job-ad language.¹⁸ This hypothesis does not necessarily distinguish between taste and statistical discrimination but rather just tests whether employers who do not want to hire older workers use stereotypes in job ads to facilitate their discrimination.¹⁹

A second hypothesis is more closely related to statistical discrimination. Different jobs may have different requirements, which could be stated in job ads. However, employers may hold stereotypes about older job applicants' abilities to meet these job requirements—for example, assuming that older workers are less likely to be able to do heavy lifting. In this case, employers posting such ads and offering fewer callbacks to older workers would be engaging in pure statistical discrimination.

While economists are interested in the nature of discriminatory behavior, both statistical and taste discrimination are illegal under US law. Equal Employment Opportunity Commission regulations state the following: "An employer may not base hiring decisions on stereotypes and assumptions about a person's race, color, religion, sex (including pregnancy), national

¹⁷ In legal cases, the most compelling data on hiring discrimination comes from comparing hiring rates of the group in question (older workers, in our case) relative to the applicant pool. In the absence of data on applicants, the analysis of a firm's workforce relative to the age structure of the relevant workforce in the population is sometimes used, but such analyses pose a greater challenge to establishing evidence consistent with age discrimination.

¹⁸ One might be concerned that different people in an organization write the ads and make the hiring decisions. However, the job board we use is focused on smaller employers, and our experimental protocol sharpens this focus because we do not use ads from larger companies that direct job seekers to their online job application sites (see Neumark, Burn, and Button 2019).

¹⁹ Taste discrimination is discrimination that occurs because employers, employees, or customers have animus against, or a distaste for, the group in question. Statistical discrimination is defined as using actual or perceived group-level differences—such as stereotypes—to make inferences about an individual from the group and hence treat that individual differently. See Neumark (2018) for additional discussion.

origin, age (40 or older), disability or genetic information.”²⁰ The regulations do not refer to whether the stereotypes are correct (i.e., right on average), although from an efficiency perspective economists would likely be more concerned about incorrect stereotypes.

While a justification for statistical discrimination may simply be that it is unfair, labor economics research shows that statistical discrimination can generate inefficiencies when workers’ human capital investments respond to this discrimination (see the seminal 1983 paper by Lundberg and Startz). For example, in the context of aging, an assumption like “older workers do not understand new technology” can deter investments by older workers in acquiring new technological skills. Of course, age stereotypes can also be inefficient if they are wrong, and there is ample evidence that simple assumptions about declines in workers’ skills as they get older are sometimes incorrect (for recent evidence and a review, see Börsch-Supan, Hunkler, and Weiss 2021).

A somewhat different and more complicated question is whether job requirements reflected in the stereotyped language in job ads, to the extent that they result in less hiring of older workers, are legal. Legality generally requires an employer to show that these requirements are based on a reasonable factor other than age (RFOA), even if that factor is correlated with age.²¹ In other words, a job requirement that is associated with less hiring of older workers is not necessarily illegal. Our evidence does not speak to the potential legality of job requirements that reflect age stereotypes. However, evidence that such job requirements are associated with hiring discrimination against older workers would prompt important questions about the validity of these job requirements, more so if we think the first hypothesis—that employers put these requirements in job ads to discourage older workers from applying—has some validity.

We do not necessarily know—nor do we need to take a stand—on why employers discriminate on the basis of age. They may want to avoid older workers because of taste-based discrimination or because of statistical discrimination. The potential implications for the observed relationship between stereotyped language and hiring are the same.

V. Methods

An essential task in this paper is to classify job ads by the age stereotypes that appear in their text. To do this, we scrape the text and use language-processing software and algorithms to identify language related to age stereotypes and to

²⁰ See <http://www1.eeoc.gov/laws/practices/index.cfm?renderforprint=1> (viewed September 15, 2019).

²¹ An RFOA is defined as “a non-age factor that is objectively reasonable when viewed from the position of a prudent employer mindful of its responsibilities under the ADEA under like circumstances” (see <https://www.federalregister.gov/documents/2012/03/30/2012-5896/disparate-impact-and-reasonable-factors-other-than-age-under-the-age-discrimination-in-employment>; viewed September 15, 2019).

quantify the strength of the relationships.²² We then use this information to test whether employers who use language in their job ads that is related to stereotypes of older workers are less likely to hire older workers—as captured in the experimental results.

Our strategy was to specify the relationships between job-ad language and age stereotypes *ex ante*, prior to doing any analysis of which job-ad language predicts measured discrimination, as well as to make the identification of which phrases from job ads predict discrimination mechanical. This dual strategy was intended to avoid (i) cherry-picking phrases from job ads that predict age discrimination, (ii) specification search to emphasize results suggesting that stereotyped phrases are associated with discrimination, and (iii) *ex post* rationalization of the results (finding which phrases in the job ads predict discrimination and then searching for age stereotypes related to these phrases).²³

Our first step was to identify common age stereotypes from the research literature in industrial psychology and related fields. Second, we used computer science methods for measuring semantic similarity in text data to identify and code words and phrases in the job ads that are related to specific age stereotypes (Mikolov et al. 2013a, 2013b). Third, for each job ad we used all of the phrases in the job ad to calculate the job-ad-specific distribution of semantic similarity scores for each stereotype. We quantified the usage of stereotyped language across ads for each stereotype on the basis of the 95th percentile of the distribution for each stereotype in each ad.²⁴ Finally, we regressed a dummy variable for observing age discrimination in our experiment on the 95th percentiles of each ad's similarity score distributions—for all of the stereotypes simultaneously. If we found a positive effect of the 95th percentile for a particular stereotype, the implication was that job-ad language related to that stereotype predicts hiring discrimination against older workers.²⁵ We explain these steps in the following subsections.

²² Appendix C (apps. A–C are available online) provides more details, references to computer code, etc.

²³ This strategy is reflected in our analysis using the complete Wikipedia corpus, described below. The additional analyses we present were suggested by reviewers or the editor, which of course happened after our initial work.

²⁴ For example, consider two job ads. The first has many phrases about physical ability (a negative stereotype applied to older workers), and the second does not. Then the 95th percentile of the distribution of similarity scores to the physical ability stereotype will be much higher for the first ad.

²⁵ An alternative procedure we explored was to use machine learning methods (elastic net) to identify the words and phrases from the job ads that predict age discrimination in hiring and to analyze statistically whether the selected words and phrases that predict age discrimination are related to age stereotypes, based on the semantic similarity scores. The results were qualitatively similar but much more complicated to present, explain, and interpret. (An earlier version of the paper with these results is available on request.) We are grateful to an anonymous reader of this paper for suggesting the simpler approach.

A. Identifying Stereotypes of Older Workers

We conducted a detailed review of the industrial psychology, communications, and related literature to identify age stereotypes that this research identifies as applying to workers in their 50s and 60s. We relied on studies that were more likely to cover the cohorts covered by the data in Neumark, Burn, and Button (2019), since age stereotypes may change over time (Gordon and Arvey 2004); we avoided studies published before the 1980s and studies of non-Western countries. We reviewed an extensive set of literature reviews and meta-analyses to identify the relevant studies, but we drew our stereotypes from papers that tested for stereotypes rather than papers that simply reported or aggregated the evidence on stereotypes from other studies.

We compiled lists of the stereotypes that these studies identified as applying to older workers. Since studies often have similar stereotypes but phrase them differently, we grouped very similar stereotypes into aggregate categories in a similar manner to the literature review and meta-analysis papers (e.g., Posthuma and Campion 2007).²⁶ To focus the analysis on stereotypes that research agrees on, we included a stereotype in our analysis only if at least two studies confirmed the stereotype.

This process led to 17 stereotypes of older workers, listed in tables 1–3, corresponding to stereotypes related to health, personality, and skills. Of these 17 stereotypes, 11 (including all of the health-related stereotypes) are negative—lower ability to learn, less adaptable, less attractive, worse communication skills, less physically able, less productive, worse with technology, less creative, worse memory, hard of hearing, and negative personality—and six are positive (more productive, dependable, careful, more experienced, better communication skills, and warm personality). Note that three pairs are contradictory: worse/better communication skills, warm/negative personality, and less/more productive. Our empirical analysis provides evidence on the effects of these age-related stereotypes in either direction, which is informative about the net effect of these related stereotypes—in favor of or against older workers.

B. Matching Stereotypes to Words and Phrases in the Job Ads

The most complex part of our research is the machine learning methods to identify words and phrases in the job ads that are related to the 17 stereotypes. The complication is that we do not expect age stereotypes to be expressed in the job ads precisely as they are in the research literature. Rather, there are many words and phrases that are potentially related to these 17 stereotypes, and the strength of their associations with age stereotypes can vary.

²⁶ For example, within the aggregate category of “less adaptable” we include “resistant to change” (McGregor and Gray 2002; Weiss and Maurer 2004), “adapt less well to change” (Warr and Pennington 1993), and “[less] flexibility” (Levin 1988).

Table 1
Stereotypes about Older Workers' Health

Aggregate Stereotype	Phrasing	Source
Less attractive	"wrinkled," "unattractive," "not neat"	Kite, Deaux, and Miele 1991
	"less attractive"	Levin 1988
	"worse -looking when older"	Zepelin, Sills, and Heath 1987
Hard of hearing	"hard of hearing"	Kite, Deaux, and Miele 1991
	"worse hearing," "think people speak too softly," "frustrated when not hearing," "think other people speak too fast," "often ask others to repeat"	Ryan et al. 1992
	"worse hearing"	Hummert, Gartska, and Shaner 1995
Worse memory	"worse memory"	Hendrick et al. 1988
	"worse memory"	Ryan 1992
	"worse memory"	Ryan and Kwong See 1993
	"worse memory"	Hummert, Gartska, and Shaner 1995
Less physically able	"lower physical capacity"	Kroon et al. 2016 (p. 16)
	"[worse] physical capability and health"	Van Dalen, Henkens, and Schippers 2009 (p. 21)
	"sedentary," "physically handicapped," "slow moving," "sick," "shaky hands," "fragile," "poor posture"	Schmidt and Boland 1986
	"less qualified for a physically demanding job"	Finkelstein, Burke, and Raju 1995
	"tired," "scared of becoming sick or incompetent"	Hummert et al. 1994
	"[lower] activity," "[less] energy," "[worse] health," "[less] speed"	Levin 1988 (p. 142)
	"less physically active," "unhealthy," "moves slowly"	Kite, Deaux, and Miele 1991
	"worse psychomotor speed"	Hendrick et al. 1988

Our computational linguistics methods consisted of two steps. First, we used machine learning to calibrate a model to identify the semantic similarity between words and phrases. In particular, we used machine learning to train a model using textual data from English-language Wikipedia.²⁷ The

²⁷ We use the English Wikipedia corpus as of November 3, 2017. This included 5.4 million articles (see <https://dumps.wikimedia.org/enwiki/>; viewed November 3, 2017). As is standard in the neural networks literature, we divide each Wikipedia article into paragraphs (Adafre and De Rijke 2006). We further split the paragraphs into single sentences. Each sentence and paragraph is used as a separate document in the machine learning algorithm. The intuition is that sentences can provide information on closer relationships between words, like "ice" and "cold," while paragraphs

model has a structure that relates semantic similarities among the 885,424 words used in the job ads based on their usage in Wikipedia articles.²⁸ Second, we used this Wikipedia model to calculate the similarity between the 17 stereotypes and phrases in the job ads. We now turn to a more detailed explanation of our methods.

In the first step, we trained the model using the entirety of English-language Wikipedia. The method uses neural networks, which are trained to reconstruct linguistic contexts of words. These neural networks take what would otherwise appear to be a jumble of words and sort them such that words used in similar contexts, as measured by Wikipedia, are placed closer together in a vector space. We used an algorithm called word2vec (Mikolov et al. 2013a, 2013b) to identify the similarity of two words using the context in which the words appear.²⁹ The word2vec algorithm employs a continuous “bag of words” algorithm to use the context of a word’s usage to predict other related words. The model produces a vector space where each unique word from Wikipedia is given a corresponding vector in a vector space. Words that are used more similarly to each other are located closer together in the vector space. This vector space is the mathematical representation of the relationships between these words and can be used to construct a numerical measure of the distance between any two words.

The structure of the word2vec neural network begins with the inputs (we use as our main corpus the entirety of English-language Wikipedia). It then uses a series of linear projection functions (also known as hidden layers because the researcher does not observe them) to transform the textual data into a vector space. These hidden layers sort the inputted text-based data to identify relationships between words in the inputted text, capturing more complex relationships between words in the texts with each layer. To identify relationships in the data, the model aggregates the data and shrinks the dimensions of the textual data. Each layer takes in a series of inputs from the previous layer and projects the data onto the next layer, reducing the dimensionality of the vector without losing valuable information. These projections are linear functions that weight all of the inputs to produce an output.

are needed for more general relationships, like “ice” and “Antarctica,” which are related but might be less likely to appear in the same sentence.

²⁸ Note that in the English language there are fewer than 885,424 words. For example, the *Oxford English Dictionary* (2nd ed.) includes 171,476 words in current use (<https://www.lexico.com/explore/how-many-words-are-there-in-the-english-language>; viewed April 30, 2022). But the job ads include names, places, misspellings, verb conjugations, etc.

²⁹ Our application of the word2vec algorithm is taken from <https://radimrehurek.com/gensim/models/word2vec.html> (viewed September 15, 2019). Readers interested in learning more about this method are directed to <http://kavita-ganesan.com/gensim-word2vec-tutorial-starter-code/#.XOthPIhKiUl> (viewed September 15, 2019) for an overview of the implementation of the word2vec algorithm and alternative applications.

Table 2
Stereotypes about Older Workers' Personality

Aggregate Stereotype	Phrasing	Source
Less adaptable	"[less] flexible in doing different tasks," "[less likely to] try new approaches"	AARP 2000 (p. 6)
	"occupationally flexible"	Karpinska, Henkens, and Schippers 2013
	"[more] flexibility"	Levin 1988 (p. 142)
	"[less likely to] adapt to change," "[less likely to] grasp new ideas"	Lyon and Pollard 1997 (p. 252)
	"older workers are less flexible than younger workers"	McCann and Keaton 2013
	"resistant to change"	McGregor and Gray 2002
	"find difficult to change," "old-fashioned"	Schmidt and Bolland 1986
	"adapt less well to change," "are less able to grasp new ideas"	Warr and Pennington 1993 (p. 89)
	"resistant to change"	Weiss and Maurer 2004
	"talks of past," "focuses away from future toward past"	Kite, Deaux, and Miele 1991
	"less flexible," "more old-fashioned"	Stewart and Ryan 1982
	"think before they act"	Lyon and Pollard 1997 (p. 251)
	"older workers are more cautious than younger workers"	McCann and Keaton 2013
Careful	"cautiousness," "self-discipline"	Truxillo et al. 2012 (p. 2623)
	"think before they act"	Warr and Pennington 1993 (p. 89)
	"better practical judgment," "better common sense"	Hendrick et al. 1988
	"[lower] creativity"	Levin 1988 (p. 142)
	"[lower] creativity"	Van Dalen, Henkens, and Schippers 2009 (p. 21)

Dependable	“loyal”	AARP 2000 (p. 6)
	“[more] stability”	Crew 1984 (p. 433)
	“more reliable,” “committed to the organization”	Van Dalen, Henkens, and Schippers 2009 (p. 21)
	“stable”	Finkelstein, Burke, and Raju 1995
	“trustworthy,” “reliability,” “commitment”	Kroon et al. 2016 (p. 16)
	“are loyal to the organization”	Lyon and Pollard 1997 (p. 251)
	“reliability,” “loyalty,” “job commitment”	McGregor and Gray 2002
	“loyal to the company,” “are reliable”	Pitt-Catsouphes et al. 2007 (p. 8)
	“more loyal to the organization,” “more reliable”	Warr and Pennington 1993 (p. 89)
	“more stable”	Singer 1986
Negative personality	“more trustworthy”	Stewart and Ryan 1982
	“dejected,” “poor,” “hopeless,” “unhappy,” “lonely,” “insecure,”	Kite, Deaux, and Miele 1991
	“complains a lot,” “grouchy,” “critical,” “miserly,”	
	“[less] pleasantness”	Levin 1988 (p. 143)
	“ill-tempered,” “bitter,” “demanding,” “complaining,” “annoying,”	Schmidt and Boland 1986
	“humorless,” “selfish,” “prejudiced,” “suspicious of strangers,”	
	“easily upset,” “miserly,” “snobbish”	
	“[less] friendliness,” “[less] cheerfulness”	Truxillo et al. 2012 (p. 2623)
	“warm,” “good-natured,” “benevolent,” “amicable”	Krings, Sczesney, and Kluge 2011
	“warm personality”	Kroon et al. 2016 (p. 16)
Warm personality	“more conscientious”	Warr and Pennington 1993 (p. 89)
	“warm”	Fiske et al. 2002

Table 3
Stereotypes about Older Workers' Skills

Aggregate Stereotype	Phrasing	Source
Lower ability to learn	"will [not] participate in training programs"	AARP 2000 (p. 6)
	"learn new techniques," "personal development"	Armstrong-Stassen and Schlosser 2008
	"[less] potential for development"	Crew 1984 (p. 433)
	"lack willingness to be trained"	Van Dalen, Henkens, and Schippers 2009 (p. 21)
	"training more appropriate for younger workers"	Dedrick and Dobbins 1991 (p. 373)
	"[less] ability and willingness to learn"	Kroon et al. 2016 (p. 16)
	"[less likely to] want to be trained"	Lyon and Pollard 1997 (p. 252)
	"less interest in learning"	Maurer et al. 2008
	"learn less quickly," "are less interested in being trained"	Warr and Pennington 1993 (p. 89)
	"less potential for development"	Finkelstein, Burke, and Raju 1995
Better communication skills	"lower potential for development"	Singer 1986
	"[better] interpersonal skills"	Crew 1984 (p. 433)
	"better social skills"	Van Dalen, Henkens, and Schippers 2009 (p. 21)
	"more interpersonally skilled"	Kroon et al. 2016 (p. 16)
Worse communication skills	"sincere when talking," "tells more enjoyable stories"	Ryan et al. 1992
	"less interpersonally skilled"	Finkelstein and Burke 1998 (p. 331)
	"unable to communicate"	Schmidt and Boland 1986
	"worse interpersonal skills"	Singer 1986
	"talks slowly," "less sociable," "has few friends"	Kite, Deaux, and Meile 1991
	"worse conversational skills," "hard to understand when noisy," "lose track of who said what," "lose track of topic," "lose track of what talked about," "hard to speak if pressed for time," "use fewer difficult words," "recognize meanings of fewer words"	Ryan et al. 1992
	"less outgoing," "quieter voice," "more hoarse"	Stewart and Ryan 1982
	"solid experience"	AARP 2000 (p. 6)
	"[more] experience"	Finkelstein, Higgins, and Clancy 2000
	"[more] experience"	Finkelstein, Ryan, and King 2013
More experienced	"have useful experience"	Lyon and Pollard 1997 (p. 251)
	"having more experience which is useful in the job"	Warr and Pennington 1993 (p. 89)

Table 3 (Continued)

Aggregate Stereotype	Phrasing	Source
More productive	"strong work ethic"	Pitt-Catsouphe et al. 2007 (p. 8)
	"working harder"	Warr and Pennington 1993 (p. 89)
Less productive	"[lower] performance capacity"	Crew 1984 (p. 433)
	"attributed low performance more to the stable factor of lack of ability when the subordinate was old"	Dedrick and Dobbins 1991 (p. 368)
	"less economically beneficial"	Finkelstein and Burke 1998 (p. 331)
	"high performance rating is positively related with youth"	Lawrence 1988 (p. 328)
	"[less] competence"	Levin 1988 (p. 142)
Worse with technology	"younger workers are seen as having higher performance capacity"	Singer 1986 (p. 691)
	"[less likely to] understand new technologies," "[less likely to] learn new technologies," "[less] comfortable with new technologies"	AARP 2000 (p. 6)
	"lack capacity to deal with new technologies"	Van Dalen, Henkens, and Schippers 2009 (p. 21)
	"[less] technological competence," "[less] technological adaptability"	Kroon et al. 2016 (p. 16)
	"[less likely to] accept new technology"	Lyon and Pollard 1997 (p. 252)
	"older workers adapt to new technology slower than younger workers," "younger workers are less fearful of technology than older workers"	McCann and Keaton 2013
	"problems with technology"	McGregor and Gray 2002
	"less readily accept the introduction of new technology"	Warr and Pennington 1993 (p. 89)

Each linear projection function has a series of weights that transform the data and a constant (known as the bias) that shifts the projection to improve prediction.³⁰

As the data work their way through the model, the words entered in the input phase are shifted and sorted such that words that are semantically similar

³⁰ For simplicity, imagine a neural network with two dimensions (rather than the actual 200 dimension vector space we use). y is the output of the hidden layer, which is a cardinal number such that two words closer together in meaning based on their usage in Wikipedia will have numbers closer together. y is a linear function of dummy variables for every word in the layer (x), with weights and a bias correction that allows the projection function to shift up or down to improve the predictive power of the model: $y = w_1x_1 + w_2x_2 + w_3x_3 + b$. The bias correction is (b), and the weights (w) are the coefficients of the model.

to each other are situated closer to each other in the output vector space. Each vector in the vector space acts as an address for a word, allowing us to determine the similarity between any two words based on how far they are from each other, using a numerical representation of the distance between them—the “cosine similarity score” (hereafter, CS score)—defined below.³¹

To analyze semantic similarity with massive databases like Wikipedia, the recommended vector size is between 100 and 200 nodes. The more nodes, the more precise the model will be. We picked 200 nodes to increase precision in the measurement of semantic similarity.³² Hence, the actual neural network we construct takes as its input an $885,424 \times 1$ vector containing all of the words and projects it onto an output matrix that is $885,424 \times 200$.³³ We use this matrix—the neural network created by our word2vec algorithm—to calculate the semantic similarity of two words based on the CS score (defined below).³⁴

Once we estimate the vector space, we use these CS scores to identify the words in the job ads with usage (in Wikipedia) that is highly related to the usage (again, in Wikipedia) of the age stereotypes.³⁵ However, to this point

³¹ Figure A1 (figs. A1–A4, B1, B2, C1–C3 are available online) provides an illustration of how the word2vec algorithm operates. In this case, there are five inputs that are closely related, hence (hypothetically) belonging to a single layer. The word2vec algorithm takes the vector of input words and projects them onto an output vector. The output vector is ordered such that words that are more closely related to each other are placed closer to each other (e.g., “muscle” is closer to “athlete” than to “carry,” based on usage in Wikipedia). This example features a 5×1 vector projected onto a 5×1 output vector. There is a total of five words and only one node (the second dimension of the output vector) to define the context.

³² Pennington, Socher, and Manning (2014) show that there is a considerable gain in the accuracy from 100 to 200 nodes, but after 200 the gains are very marginal (see fig. 2 of Pennington, Socher, and Manning 2014).

³³ In our word2vec algorithm, the creation of this neural network begins by working from the input layer to the output to determine the optimal weights and bias in each layer of the network (“forward propagation”). This step consists of estimating the probability that a word is between a set of other words. We select optimal weights and bias to minimize the errors of these predictions. But when using only forward propagation, the estimated output can have a high error rate. To improve the estimation, we update the biases and weights based on the error rate in the model’s prediction using a process known as backward propagation. This process of using both forward and backward propagation iterations is counted as a training iteration. For our purposes, we use five training iterations of the word2vec algorithm (the default setting in the word2vec package). After the five training iterations, we have fully calibrated the neural network and populated the vector space. Our final vector space contains one row for each of the 885,424 words used on the job ads and 200 columns containing the estimated weights from the linear projection functions.

³⁴ For more details about cosine similarity, semantic similarity, and these kinds of models, see Clark (2015) and Jurafsky and Martin (2017).

³⁵ Note that there are three pairs of stereotypes that are mirror images of each other: worse/better communication skills, warm/negative personality, and less/more productive. For these pairs, we just combine the stereotypes: communication

our explanation (and the example in fig. A1) have been based on single words. Because a single word may often fail to contain enough information about the association with a stereotype (which are typically expressed in multiple words), we instead use three-word phrases from the job ads in our analysis, or “trigrams.” We create these trigrams by removing words such as “the,” “and,” or “a”—so-called stopping words in language processing—and then creating all trigrams from the remaining words. The trigrams are all sets of three consecutive words, excluding stopping words. We retain the stereotypes as the words in which they are expressed in the first column of tables 1–3 after we remove the words indicating the direction of the stereotype, such as “more” or “less.”

Then we calculate the CS score between each stereotype and every trigram used in the entire set of job ads. Because the word2vec model is created using single words, we have estimated weights only for single words. To calculate the CS score between stereotypes and trigrams, we recover the weights applied to the hidden layers in the network that correspond to the word in question, apply these weights to generate new weights for the trigrams and stereotypes, and then use the vectors of these new weights to calculate the CS score.

We first estimate the vector corresponding to the three words in the trigram (or the words in a stereotype), adding the weights element by element for each word.³⁶ For example, if the model uses two hidden layers, producing two weights for each word of three words in the trigram “able lift lbs,” then the total vector of weights of the trigram is computed as

$$\text{able lift lbs} = \begin{bmatrix} 0.3 \\ 0.2 \end{bmatrix} + \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.5 \end{bmatrix}. \quad (8)$$

We then estimate the CS score between these vectors for every trigram from the job ads and every stereotype. The CS score is defined as

$$\text{CS}(\text{trigram}, \text{stereotype}) = \frac{\text{dot product}(\text{trigram}, \text{stereotype})}{\| \text{trigram} \| \| \text{stereotype} \|}, \quad (9)$$

skills, personality, and productive. Thus, we end up looking at CS scores with these 14 stereotypes. When we discuss the results below, we explicitly consider the evidence on these ambiguous stereotypes.

³⁶ This procedure is derived from Mikolov, Yih, and Zweig (2013). They demonstrate that the relationships between words captured by the methods we use also capture relationships between small numbers of words (their focus is on pairs), based on addition or subtraction of the vectors corresponding to these words. As a prime example, the representation of the word “queen” can be roughly recovered from the representations of “king,” “man,” and “woman”—i.e., $\text{queen} \approx \text{king} - \text{man} + \text{woman}$.

where “trigram” and “stereotype” in the equation refer to the vectors of weights.³⁷

The CS score varies between -1 and 1 . A score of -1 means the words never appear in the same sentences or paragraphs in Wikipedia. As the CS score increases, the usage of the words becomes more similar; that is, they are used more often in the same sentences or paragraphs, suggesting that they are often used to discuss the same topic. This is what the literature defines as greater semantic similarity. If the words coincide perfectly, the CS score equals 1 . As an example, fig. A2 shows the distribution of CS scores of all trigrams with a particular stereotype (communication skills); the distribution is centered above zero, which makes sense because we are looking at text from job ads. To provide some examples, trigrams at the lower end of the distribution are highly unrelated. These include “christmas season near” and “hotel near seattle” (both with scores of -0.3). Trigrams with scores close to 0.0 include “every Sunday pm” and “work year round.” Trigrams at the top of the distribution with scores of 1.0 include “excellent communication skills” and “prioritizing skills communication.”

For each job ad, we use these CS scores to calculate the distribution of semantic similarity for all three-word phrases and all stereotypes. To illustrate, consider the job ad in figure 2. The ad contains phrases that, on the surface, could be related to age stereotypes, including, for example, “computer savvy,” “experience preferred,” “energetic,” and “customer friendly.” Figure 3 displays the distributions of the CS scores for each phrase (trigram) in the ad, with the communication skills, physical ability, and technology stereotypes. There is a wide range of CS scores in this job ad, although they are more related than unrelated (lying almost entirely above zero). The distributions are skewed with a long upper tail, indicating that, although rare, the job ad does contain highly related trigrams.

Our analysis requires a summary measure of the distribution of CS scores in each job ad for each stereotype, to quantify the usage of stereotyped language (for each stereotype) across job ads. We plotted the distributions of CS scores at the median, the 75th percentile, the 95th percentile, and the maximum. Figure 4 shows these for the same three stereotypes used in figure 3. The histograms in figure 4 provide a sound rationale for using the 95th percentile. The mass of the distributions is much lower using the median (or the 75th percentile). This is not surprising. If a job ad contains only a few stereotyped phrases, then the phrase with the median CS score in the ad, for a given stereotype, is likely quite unrelated to that stereotype. If we used lower percentiles, such as the median, as our measure of ageist sentiment in a job ad, we would be using variation in the language across job ads that is largely unrelated to the stereotypes we are studying. In addition, figure 4 shows that if we used the median (or the 75th percentile), we would

³⁷ The $\|$ notation indicates the Euclidean norm—e.g., $\|[x, y]^T\| = (x^2 + y^2)^{1/2}$.

Part Time Insides Sales/Customer Service Rep

[Company Name] is currently seeking Part -Time Customer Service Representative/Inside Sales Consultant to work in our beautiful Los Angeles showroom located on Westwood Blvd. This person will be responsible for maintaining, growing and acquiring new revenue from the existing Fashion customer base, as well as, generating new business. The ideal candidate will be operationally minded and have excellent sales abilities. Candidate must be reliable, energetic, customer-friendly, detail-oriented, and possess excellent time management skills.

Our Mission is to provide an unparalleled experience by fostering an environment that inspires teamwork and unrivaled passion for exceptional service.

[Company Name] is the only furniture rental company that solely services the California market. Based in San Diego, [Company Name] serves all of California, from San Diego, through The Bay Area. Please visit our website for more information [Company Website]

Essential duties and responsibilities include:

- Manage the day-to-day relationships with existing customer base and new prospects in partnership with field counterparts in assigned territory.
- Take orders and inquiries via incoming calls, walk in appointments, and email.
- Provide excellent customer service.
- Perform order and data entry with precision in a given time frame.
- Assist with residential and home staging furniture rentals.
- Assist outside sales reps with client issues & various projects.
- Provide support for internal and external customers.
- Assist in producing leads for new business
- Show a willingness to do whatever is needed to support the team

The position is heavily customer service oriented, and includes order processing. You will work with customers both over the phone and in person.

Requirements:

- Must possess excellent telephone etiquette, communication and organizational skills
- Candidate must be self motivated and be able to work independently.
- Applicants should have exceptional verbal and written skills, specifically the ability to coordinate multiple activities for numerous accounts.
- Candidate must maintain a willingness and ability to learn and understand our products and procedures.
- Effective time management
- Must be computer savvy; experience with MS Outlook and Office required
- 2-4 years of inside sales/customer service experience preferred.
- Property Management, Design/Merchandising, and Retail Sales experience a plus
- Must be available to work Saturdays

Our company is a stable, employee oriented organization, providing a fun, team oriented atmosphere.

FIG. 2.—Text of a job ad. The text is drawn from an actual job ad applied to in Neumark, Burn, and Button (2019), posted in June 2015. Company name and contact details have been removed.

not pick up much range in the language across ads. This is also not surprising for the same reason; the phrases with, for example, median CS scores are more likely to be generic phrases that do not vary much across job ads.

Since higher CS scores indicate a stronger relation to the stereotype, selecting a higher percentile of the CS score distribution ensures that the phrases being analyzed are more related to the stereotypes. On the other hand, if we use the maximum, we get what appears to be a good deal more noise (especially for communication). We would expect this because we are looking at extremes of the distribution, and language in the extreme upper tail but with different CS scores may not imply any real differences in

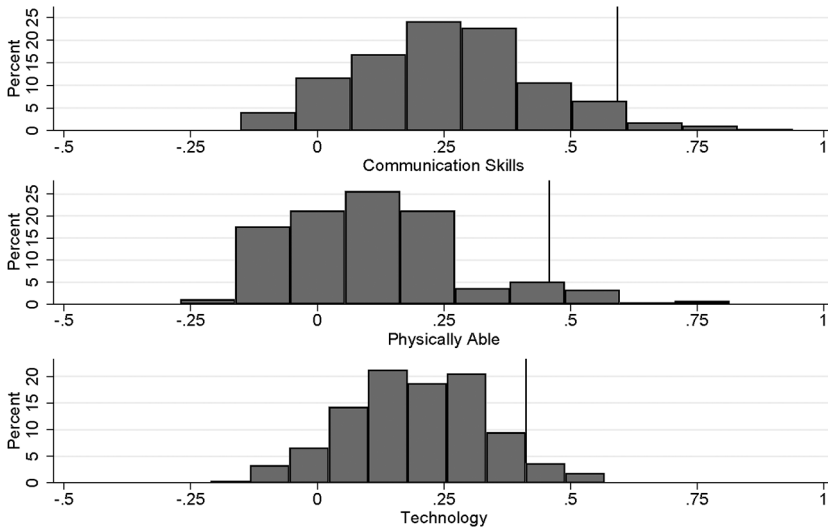


FIG. 3.—Distributions of CS scores within example ad (from fig. 2). Based on the text from the job ad in figure 2, these histograms plot the distributions of CS scores for all trigrams in the example job ad, for three stereotypes. The solid line in each graph indicates the 95th percentile.

behavior. Therefore, we chose to use the 95th percentile of the distributions of the CS scores to measure how stereotyped the language in a job ad is to quantify differences in the usage of stereotyped language across job ads.³⁸

To provide more details and examples, in figure 5 we display the distributions—for each stereotype related to health—of the 95th percentiles of the CS score computed from each job ad.³⁹ The figure displays this information for all occupations combined and then each occupation separately. The figure shows that the trigrams in job ads are fairly weakly related to hearing and memory but that ads with language strongly related to physical ability are relatively common. For this latter stereotype, the distributions in all

³⁸ We explore robustness to this choice in app. B, discussed below.

³⁹ As noted earlier, we combine the stereotypes that the literature indicates could be both positively or negatively associated with older workers (e.g., “warm personality” and “negative personality” becoming “personality”). The CS scores do not differentiate between the sentiment of the words. For example, phrases such as “good personality positive” and “bad personality negative” will both have a positive CS score with personality. We let the evidence on the association between stereotyped language and the experimental measure of age discrimination tell us whether, on net, language associated with these ambiguous stereotypes is associated with higher or lower hiring of older applicants.

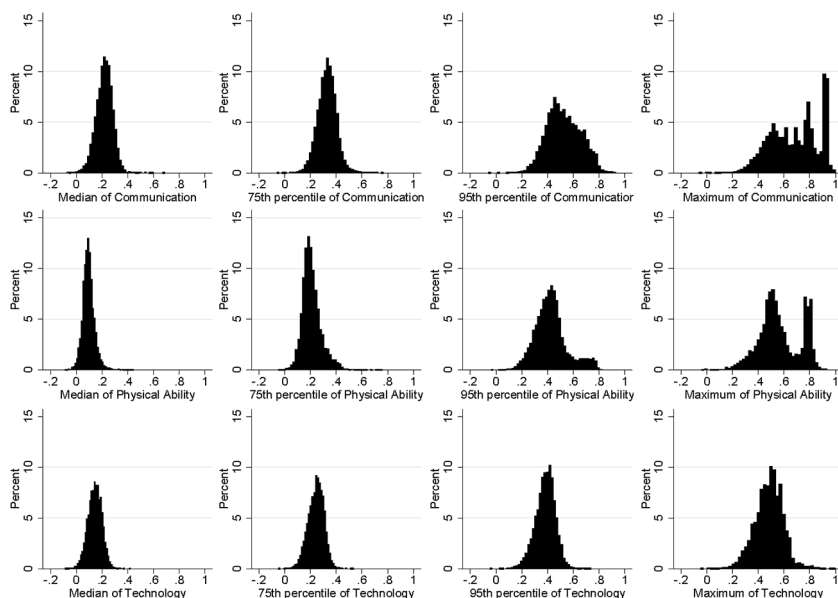


FIG. 4.—Alternative percentile distributions of CS scores of job-ad trigrams with select stereotypes. Data come from the job ads collected in Neumark, Burn, and Button (2019). The distributions are for all of the ads in our sample. Units on horizontal axes are the CS scores for the indicated percentiles/values of the distributions. A complete set of figures is available on request.

occupations feature a large mass of ads with trigrams in the range of 0.6 or higher.⁴⁰

⁴⁰ Figures A3 and A4 show the same information for stereotypes related to personality and related to skills, respectively. For personality, the distributions of the 95th percentiles of the CS scores are more normally (or at least symmetrically) distributed than those in fig. 5. The job ads do not include phrases highly related to many of the stereotypes, with the median of the 95th percentile in the 0.2–0.4 range, and the distributions do not have the large upper tails we saw for physical ability. Still, there is some variation apparent by both stereotype and occupation. For example, the distributions for careful in the security guard and the janitor ads are shifted notably to the right (and the distributions for this stereotype are furthest to the right for administrative assistant ads as well, although not as markedly). For skills, we find that job ads contain a higher frequency of trigrams related to some skill-related stereotypes. Job-ad language strongly related to ability to learn and communication skills is much more common than for the other three stereotypes in this figure and for most of the stereotypes shown in the other two figures (figs. 5, A3). However, the upper tails are not as extreme as for physical ability. Phrases in job ads strongly related to the technology stereotype are also more common (although not as pronounced); for all occupations except janitors, the medians are closer to 0.4. Phrases strongly related to experience and productivity are less common, with medians of the distributions of the 95th percentile scores in the 0.2–0.3 range and sometimes below 0.2.

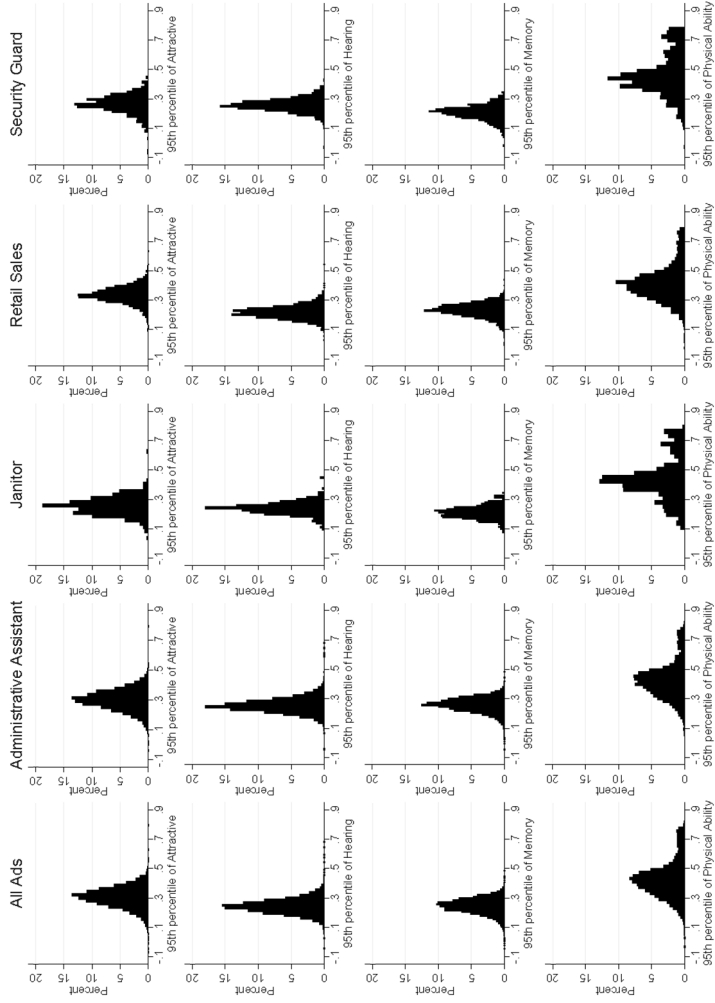


FIG. 5.—Distributions of the 95th percentiles of CS scores for stereotypes related to health. Data come from the job ads collected in Neumark, Burn, and Button (2019). Each panel plots the distribution of CS scores at the 95th percentile for the job ads with each stereotype related to health. The first column contains the distribution of all of the ads in our sample. The remaining columns disaggregate the job ads by occupation.

C. Testing Which Stereotypes Predict Callback Differences by Age

Our next step is to use our job-ad-level measures of the semantic similarity between language in the ads and age stereotypes to estimate the relationships between age-stereotyped language in job ads and the likelihood that older or younger applicants received callbacks.

Our data set includes all responses to the triplet of job applications sent in response to each job ad that we could match to an employer and their job advertisement.⁴¹ Our outcome (d_{exp}^i) is a dummy variable equal to 1 if the older applicant i did not receive a callback from employer (or, equivalently, job ad) j but the younger applicant did and 0 otherwise. If both applicants are called back, neither applicant is called back, or only the older applicant is called back (which is less common than the reverse case), we do not consider the outcome to reflect age discrimination and code d_{exp}^i as 0.⁴² In 76% of cases neither applicant was called back, while in 6% of cases both applicants were called back. In 11% of cases the older applicant was not called back and the younger applicant was, whereas the reverse occurred in 7% of cases.⁴³

We estimate probit models for our experimental measure of discrimination. The key independent variables are the 95th percentiles of the distributions of the CS scores of each phrase in the job ad with each stereotype.

⁴¹ We were able to match 34,260 job applications to 11,420 job advertisements, corresponding to 22,840 observations for older and middle-aged applicants. There were 4,266 applications that could not be matched to a saved job ad. The most common reason was that an ad was not saved. In some cases, the ad was saved in the incorrect format and could not be scraped. (Research assistants were instructed to save all job advertisements as an HTML file, but there were instances of advertisements being saved as a PDF or a PNG file.) In total, 87% of applications were matched to a job ad.

⁴² In theory, it is possible to impose an even stronger definition of discrimination on the data, defining discrimination as occurring if the younger applicant is called back but neither older applicant is. The challenge in using this definition is in the construction of the triplets. All triplets had one younger applicant and two older applicants, but the older applicants could either be middle-aged or older. So in some triplets the older workers will be a mixed pair, one old and one middle-aged. In these cases, the stronger definition of discrimination would require discrimination to occur against the applicant aged 49–51 and the applicant aged 64–66. However, in Neumark, Burn, and Button (2019) we generally observed stronger evidence of discrimination against older applicants than middle-aged applicants and sometimes no discrimination against middle-aged applicants. The way we define discrimination here is more informative, as it results in separate estimates for middle-aged vs. younger applicants and older vs. younger applicants. (This issue could be avoided in future studies by simply sending pairs of applicants in response to each job ad.)

⁴³ In each triplet sent to a job opening, there was one young worker and two older workers (randomly selected to be either middle-aged or old). Our unit of observation is each middle-aged or older applicant, so that each triplet produces two observations. Thus, discrimination against an older applicant is measured independently of whether the other older worker was called back.

Denote these percentiles, for job ad j and stereotype s , by P_{js}^{95} . We also control for the observable resume differences using the same control variables X_{ij} as in Neumark, Burn, and Button (2019).⁴⁴ Thus, our model is

$$\Pr[d_{\text{exp}}^{ij} = 1] = \alpha + \sum_s \beta_s P_{js}^{95} + X_{ij}\delta + \varepsilon_{ij}. \quad (10)$$

We standardize P_{js}^{95} (for each stereotype s) to have a mean of 0 and a standard deviation of 1 across all of the ads in the study. The term β_s then represents the effect of a 1 standard deviation increase in the 95th percentile of the CS score in a job ad for stereotype s .⁴⁵ The estimate of β_s then reflects both the effect of the stereotype on discrimination and how related the job-ad language is to the stereotype. A small β_s could be attributable to stereotype s not mattering much for discrimination or to a 1 standard deviation increase in similarity to stereotype s being a small increase. From the point of view of asking which stereotypes in job-ad language predict age discrimination, this combined effect is of interest. In contrast, a comparison between the estimated effects of the same absolute change in the 95th percentile for different stereotypes is of less interest, given that, in reality, job-ad language is more closely related to some stereotypes than to others.

We estimate our models at the gender-age-occupation level (e.g., women aged 49–51 applying to sales jobs) to allow the effects of stereotypes to vary across different kinds of employers and applicants. Because each job ad has two pairs of applicants—one younger applicant and one middle-aged or older applicant in each pair—we cluster the standard errors at the job-ad level.

VI. Results

We now turn to the evidence on the relationships between age stereotypes in job ads and age discrimination in hiring. Figure 6 presents a convenient summary of the results from estimates of equation (10) by age, gender, and occupation, and the full underlying estimates are reported in table 4; both show the estimated marginal effects. The top row of table 4 shows the percentage of job ads where the younger applicant received a callback but the older applicant did not—our experimental measure of discrimination. This occurred in 11% of cases, but the rate varies by occupation. For the regression estimates below the first row, the coefficients measure the marginal

⁴⁴ Resume features include city, order sent, skill level, unemployment status, template, and email domain.

⁴⁵ In table A1 (tables A1–A3, B1–B3 are available online), we provide the text of some trigrams to indicate how they differ 1 standard deviation higher in the distribution of 95th percentile CS scores. We report the text at the mean 95th percentile and the five trigrams closest to 1 standard deviation higher than this mean. This corresponds to the interpretation of β_s , although of course identification of β_s comes from variation across the entire distribution.

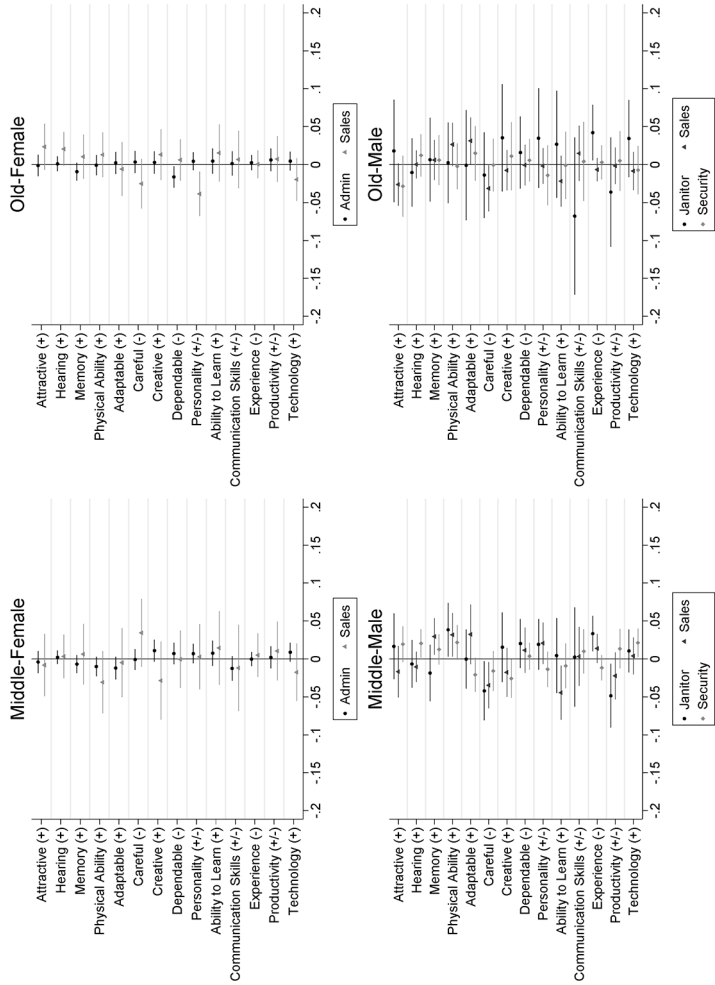


FIG. 6.—Baseline results by gender, age, and occupation. This figure reports estimated marginal effects from equation (10) and 95% confidence intervals. Predicted signs of each coefficient according to the industrial psychology literature are presented in parentheses. Positive predictions indicate that we expect to see more discrimination against older applicants, and negative predictions indicate that we expect to see less discrimination against older applicants. Standard errors are clustered at the job-ad level.

Table 4
Baseline Results by Gender, Age, and Occupation: Effects on Discrimination against Older Applicants

	Predicted Sign	Female				Male					
		Middle-Admin (1)	Middle-Sales (2)	Old-Admin (3)	Old-Sales (4)	Middle-Janitor (5)	Middle-Sales (6)	Middle-Security (7)	Old-Janitor (8)	Old-Sales (9)	Old-Security (10)
Baseline		.100	.155	.103	.140	.096	.104	.089	.148	.102	.121
discrimination Health stereotypes:											
	Attractive	-.004 (.007) [.797]	-.008 (.021) [.967]	-.001 (.007) [.893]	.023 (.015) [.455]	.017 (.022) [.714]	-.017 (.017) [.419]	.019 (.012) [.243]	.018 (.034) [.823]	-.026 (.014) [.242]	-.029 (.020) [.969]
	Hearing	.002 (.004) [.797]	.003 (.015) [.967]	.001 (.005) [.893]	.021 (.011) [.455]	-.007 (.016) [.864]	-.010 (.010) [.419]	.020* (.010) [.185]	-.010 (.023) [.823]	.000 (.009) [.983]	.012 (.014) [.969]
Memory	Positive	-.007 (.006) [.558]	.006 (.020) [.967]	-.009 (.006) [.847]	.010 (.015) [.751]	-.019 (.019) [.648]	.030* (.012) [.102]	.012 (.010) [.346]	.006 (.028) [.959]	.006 (.013) [.913]	.006 (.017) [.969]
Physical ability	Positive	-.010 (.007) [.451]	-.031 (.021) [.967]	-.001 (.007) [.893]	.013 (.015) [.751]	.038* (.018) [.119]	.032* (.015) [.102]	.022 (.012) [.185]	.002 (.027) [.978]	.026 (.015) [.242]	-.002 (.015) [.969]
Personality stereotypes:											
	Adaptable	-.012 (.008) [.451]	-.005 (.023) [.967]	.002 (.007) [.893]	-.006 (.018) [.798]	-.000 (.020) [.991]	.032 (.020) [.273]	-.021 (.011) [.185]	-.001 (.037) [.978]	.031* (.016) [.242]	.015 (.018) [.969]
Careful	Negative	-.001 (.007) [.958]	.034 (.023) [.967]	.003 (.007) [.893]	-.025 (.017) [.455]	-.042* (.020) [.119]	-.035* (.015) [.102]	-.016 (.013) [.346]	-.014 (.029) [.823]	-.031* (.015) [.242]	-.001 (.018) [.969]
Creative	Positive	.011 (.007) [.451]	-.029 (.026) [.967]	.003 (.008) [.893]	.013 (.017) [.751]	.015 (.023) [.714]	-.018 (.016) [.419]	-.026* (.013) [.185]	.035 (.036) [.763]	-.008 (.014) [.893]	.011 (.023) [.969]

Dependable	Negative	.007 (.007) [.558]	-.001 (.019) [.979]	-.016* (.007) [.322]	.006 (.014) [.798]	.020 (.016) [.602]	.011 (.015) [.522]	.004 (.009) [.677]	.016 (.024) [.823]	-.001 (.014) [.983]	.006 (.014) [.969]
Personality	Positive/negative	.007 (.006) [.558]	.003 (.022) [.968]	.004 (.006) [.893]	-.039** (.015) [.140]	.019 (.017) [.604]	.021 (.014) [.273]	-.014 (.012) [.346]	.035 (.033) [.763]	-.002 (.012) [.983]	-.014 (.020) [.969]
Skills stereotypes:											
Ability to learn	Positive	.008 (.008) [.576]	.014 (.025) [.967]	.005 (.008) [.893]	.015 (.019) [.751]	.004 (.025) [.991]	-.045* (.018) [.102]	-.009 (.015) [.579]	.027 (.036) [.823]	-.022 (.017) [.560]	-.001 (.023) [.969]
Communication skills	Positive/negative	-.013 (.008) [.451]	-.012 (.029) [.967]	.001 (.008) [.893]	.007 (.019) [.798]	.002 (.033) [.991]	.003 (.020) [.869]	.010 (.015) [.579]	-.068 (.053) [.763]	.015 (.019) [.856]	.004 (.027) [.969]
Experienced	Negative	-.000 (.005) [.958]	.005 (.015) [.967]	.002 (.005) [.893]	.000 (.009) [.959]	.033** (.012) [.070]	.014 (.010) [.273]	-.012 (.008) [.318]	.042* (.019) [.336]	-.007 (.008) [.856]	.003 (.011) [.969]
Productive	Positive/negative	.002 (.007) [.922]	.010 (.020) [.967]	.006 (.008) [.893]	.007 (.015) [.798]	-.049* (.021) [.119]	-.022 (.016) [.273]	.014 (.013) [.391]	-.036 (.037) [.763]	-.002 (.012) [.983]	.005 (.020) [.969]
Technology	Positive	.009 (.006) [.451]	-.018 (.019) [.967]	.005 (.006) [.893]	-.020 (.014) [.473]	.011 (.014) [.714]	.004 (.012) [.810]	.021* (.009) [.185]	.034 (.026) [.763]	-.009 (.013) [.884]	-.007 (.016) [.969]
<i>N</i>		6,822	986	7,321	1,856	311	1,612	954	318	1,680	932
Adjusted <i>R</i> ²		.026	.055	.026	.047	.208	.055	.164	.128	.060	.076

NOTE.—This table reports estimated marginal effects from eq. (10). Predicted signs of each coefficient are according to the industrial psychology literature (see tables 1–3). Positive predictions indicate that we expect to see more discrimination against older applicants, and negative predictions indicate that we expect to see less discrimination against older applicants. Standard errors are clustered at the job-ad level. In brackets we report *p*-values corrected for multiple hypothesis testing within a regression (i.e., a column) using the Simes procedure. Boldface type indicates statistical significance at the 5% level using conventional significant tests.

* Statistically significant at the 5% level.
 ** Statistically significant at the 1% level.

effect on the probability of age discrimination of a 1 standard deviation increase in the CS score at the 95th percentile of the job ads' distributions for that stereotype. For example, middle-aged men who applied for a job as a janitor (in col. 5) experienced age discrimination 9.6% of the time. If the job ad included language where the 95th percentile was 1 standard deviation more highly related to physical ability than the average ad, measured discrimination was 3.8 percentage points (or 40%) higher.⁴⁶

Figure 6 provides a compact way to visualize the estimates, displaying the estimated marginal effects and their 95% confidence intervals.⁴⁷ Figure 6 shows that stereotyped language in job ads is associated with hiring discrimination, especially for men. We observe significant heterogeneity across occupations in which stereotypes are associated with hiring discrimination. In some age-occupation cells, we observe only one or two significant effects, and in some cases we observe more. When we take a more aggregated view, we find that middle-aged men have the highest number of stereotypes correlated with hiring discrimination (11), followed by older men (three) and older women (two), while middle-aged women have the fewest (zero). For older women, stereotyped language related to personality appears to be most strongly correlated with differential hiring by age. Among men, we observe health (but only for middle-aged men), personality, and skill-related stereotypes all being associated with hiring discrimination.

We now discuss these results in more detail. In general, our evidence regarding the relationships between stereotyped job-ad language and discrimination

⁴⁶ Table 4 includes results from correcting for multiple hypothesis testing (p -values are in brackets). We use the Simes procedure to control the false discovery rate. The false discovery rate (e.g., at the 5% significance level) means that we are 95% confident that at least some of the rejected hypotheses are false. The more conservative approach controls the family-wise error rate, which means that we would be 95% confident that all rejected hypotheses are false. Controlling the family-wise error rate is more appropriate when there is the potential for "harm" from falsely rejecting any of the tested hypotheses. In contrast, controlling the false discovery rate is more appropriate when harm is less likely to be caused by a single true hypothesis being falsely rejected, as long as some are correctly rejected (Pike 2011). We think the latter is more appropriate to our case, as we are testing, in general, whether age stereotypes in job ads predict hiring discrimination (as opposed, say, to testing the effects of a number of interventions on a disease). We would also suggest that the need for these corrections is less compelling in our case than in some others, as we prespecified the stereotypes we study and use machine learning to measure semantic similarity. We thus do not have the issue of, e.g., reporting a subset of results from a large set of possible analyses (as discussed, e.g., in Casey, Glennerster, and Miguel 2012). The adjusted p -values are higher in all instances. None of the results for women remain significant, but the strongest results for men remain significant or marginally significant in many cases.

⁴⁷ Note that the horizontal scale is not the same in every figure—including the other versions of these figures that come later.

against older job applicants is consistent with what we would expect from the evidence on stereotypes in the industrial psychology literature.

A. Results for Stereotypes Related to Health

In figure 6 (and table 4), we find that stereotypes related to the health of older workers predict higher levels of discrimination for middle-aged men but not for women (of either age group) or older men. The differences are primarily attributable to larger point estimates of the effects for middle-aged men rather than more precise estimates. For middle-aged men, the health-stereotyped language is always associated with more discrimination.

The stereotypes that are associated with increased discrimination depend on the occupation. For retail sales and janitor positions, we find that employers with job ads that used language more highly related to physical ability more often discriminate against middle-aged men (significant at the 5% level). The estimates for middle-aged men in security jobs and older men in sales jobs are in the same direction, with slightly weaker statistical evidence (significant only at the 10% level). We expected to find this on the basis of the industrial psychology literature indicating that employers view older workers as having lower physical abilities than younger workers (see table 1). For janitor positions, job ads with phrases for which the 95th percentile of the distribution of CS scores with physical ability were 1 standard deviation higher are associated with a 3.8 percentage point increase in discrimination against middle-aged men. Similarly, for sales positions a 1 standard deviation increase in this 95th percentile is associated with a 3.2 percentage point increase in discrimination against middle-aged men.⁴⁸

For retail sales positions, we find that employers with job ads that used language more highly related to memory more often discriminate against middle-aged men. This is expected on the basis of the industrial psychology literature indicating that employers view older workers as having worse memories than younger workers (see table 1). Job ads with phrases for which the 95th percentile of the distribution of CS scores with memory were 1 standard deviation higher are associated with a 3.0 percentage point increase in discrimination against middle-aged men.⁴⁹ For security guard

⁴⁸ In table A1, we show that this 1 standard deviation increase in the CS score for physical ability is akin to changing the 95th percentile from “assistant position available” to a phrase similar in relatedness to “work preferred necessary,” “fast-paced fun,” or “required flexibility required.”

⁴⁹ This result highlights a potential challenge in adapting methods from machine learning to our context of analyzing text data. The word2vec algorithm could sometimes identify trigrams from the job ads that are not meaningfully related to our age stereotypes—which we might think of as false positives—and if these trigrams happen to predict lower relative callback rates for older job applicants, these false positives could generate bias toward concluding that job-ad language related to specific age stereotypes predicts age discrimination. The stereotype “memory” may be a

positions, we find that employers with job ads that used language more highly related to hearing more often discriminate against middle-aged men. This evidence is consistent with the industrial psychology literature indicating that employers view older workers as having worse hearing than younger workers (see table 1). Job ads with phrases for which the 95th percentile of the distribution of CS scores with hearing were 1 standard deviation higher are associated with a 2.0 percentage point increase in discrimination against middle-aged men.

B. Results for Stereotypes Related to Personality

Stereotypes related to personality appear to contribute to age-related differences for older women but not for middle-aged women; the evidence indicates that stereotyped language related to personality is associated with less discrimination against older women. For older women in administrative jobs, employers who use language in their job ads more related to dependability discriminate less against older workers. This result is in line with the predictions of the industrial psychology literature, which indicates that employers view older workers as more dependable than younger workers (see table 2). For administrative assistant job ads, those with phrases for which the 95th percentile of the distribution of CS scores with dependability were 1 standard deviation higher decrease measured discrimination against older women by 1.6 percentage points.

For older women in retail jobs, employers who use language in their job ads more related to the personality stereotype discriminate less against older workers. In the industrial psychology literature, employers' views of the personalities of older workers are mixed (see table 2); some papers suggest that employers view older workers as having worse personalities, while others provide evidence that employers view them as having warm personalities. Our results suggest that in retail sales the more positive stereotype dominates for older women. For sales associate positions, job ads with phrases for which the 95th percentile of the distribution of CS scores with personality were 1 standard deviation higher decrease measured discrimination against older women by 3.9 percentage points.

For middle-aged men applying to janitor positions and for both older and middle-aged men applying to retail sales positions, we find that employers with job ads with phrases more highly related to careful are less likely to discriminate against older workers. This result matches the industrial psychology

prime example. We find a significant correlation between using words highly related to memory and discrimination against middle-aged men. But this result could be driven in part by an ad mentioning "computers," which was identified by the algorithm as being highly related to "memory" because of the number of times "computer memory" is mentioned in Wikipedia. Therefore, we may identify a significant correlation for memory when in reality we are picking up a technology-related correlation.

literature, which indicates that employers view older workers as more careful than younger workers (see table 2). For janitor job ads, when phrases for which the 95th percentile of the distribution of CS scores with careful were 1 standard deviation higher, measured discrimination was 4.2 percentage points lower. For sales associate job ads, a 1 standard deviation increase in this 95th percentile decreases observed discrimination by 3.5 percentage points for middle-aged applicants and 3.1 percentage points for older applicants. For middle-aged men applying to security guard positions, we observe results that are at odds with the industrial psychology literature, which finds that older workers were viewed by employers as less creative (see table 2). In contrast, we find evidence that employers who use language related to creativity are less likely to discriminate against older workers. Job ads with phrases for which the 95th percentile of the distribution of CS scores with creative were 1 standard deviation higher are associated with measured discrimination being 2.6 percentage points lower. Finally, for older men in retail sales we find that language related to adaptability is associated with increased discrimination. This finding is consistent with the industrial psychology literature, which shows that older workers are viewed as less adaptable (see table 2). Job ads with phrases for which the 95th percentile of the distribution of CS scores with adaptability were 1 standard deviation higher increase observed discrimination by 3.1 percentage points.

C. Results for Stereotypes Related to Skills

For women, we do not observe any significant relationship between phrases highly related to skill-related stereotypes and observed discrimination. This is true in both administrative assistant and retail sales positions, as well as across age groups. The very small point estimates in the administrative assistant models suggest that the language may be very similar on ads where the employer discriminates and ads where the employer does not. We observe larger differences in retail sales ads, but the smaller sample size results in larger standard errors. Still, the point estimates are smaller in absolute value than the significant results we have discussed above.

Among men, we observe skill-related language being associated with discrimination against older workers in several age-occupation cells. A wider range of stereotyped language predicts discrimination for middle-aged men, and we observe significant heterogeneity across occupations. In contrast, the evidence for older men appears only in janitor positions.

In the industrial psychology literature, there is disagreement about whether employers stereotype older workers as more productive or less productive (see table 3). Thus, as for personality, this is a case where our method enables us to gauge the relative importance of the positive or negative association empirically by measuring employer behavior. Our results suggest that among the employers who are hiring janitors, the positive association between age and productivity dominates. We find that employers who use language

more highly related to productivity are less likely to discriminate against middle-aged men. Job ads with phrases for which the 95th percentile of the distribution of CS scores with productivity were 1 standard deviation higher decrease measured discrimination by 4.9 percentage points.

For middle-aged men in security guard positions, we observe our only instance of technology-related language being correlated with hiring discrimination. Despite a significant emphasis on this in the industrial psychology literature, it does not appear that the usage of language related to technology often differs between employers who discriminate against older workers and those who do not. Our estimated correlation is in the same direction as the negative stereotypes about older workers and technology suggested by the literature (see table 3), as we observe a higher rate of discrimination associated with job ads that use language highly related to technology. Job ads with phrases for which the 95th percentile of the distribution of CS scores with technology were 1 standard deviation higher increase observed discrimination against middle-aged men by 2.1 percentage points.

For middle-aged and older men applying to janitor positions, we observe significant associations between measured discrimination and job-ad language related to experience. In the industrial psychology literature, employers view older workers as more experienced, perhaps tautologically. In contrast, we find that employers with job-ad language more highly related to experience are more likely to discriminate against middle-aged and older men. Job ads with phrases for which the 95th percentile of the distribution of CS scores with experience were 1 standard deviation higher increase observed discrimination against middle-aged men by 3.3 percentage points and against older men by 4.2 percentage points.

D. Summary

Overall, our results suggest that ageist stereotypes may affect the hiring of older workers and in particular are associated with hiring discrimination against older men. Using the ageist stereotypes found in the industrial psychology literature, we show that job-ad language that is highly related to ageist stereotypes is associated with hiring discrimination for older men. The direction of these empirical correlations is generally in the same direction as predicted by the literature. Positive stereotypes are correlated with less hiring discrimination, and negative stereotypes are correlated with more discrimination. Across both men and women, we find only three instances where our results contradict the predictions of the industrial psychology literature, with negative stereotypes associated with less hiring discrimination. In contrast, in 13 cases the evidence is consistent with the industrial psychology literature.

VII. Supplemental Analyses

We now turn to alternative ways to conduct our analysis, examining how robust our results are to different choices about how to define discrimination,

the construction of phrases, and alternative corpora or methods to flag stereotyped job-ad language.⁵⁰

A. Definition of Discrimination

To this point, we have studied discrimination against older applicants, defined as a callback to younger applicants but not older applicants. It is possible that studying discrimination in favor of older applicants (against younger applicants) would detect more evidence of positive stereotypes reducing discrimination against older workers. We therefore also did analyses for this reverse outcome—when only the older applicant received a callback. For this analysis, we created separate pairs of each older applicant in the pair combined with the corresponding younger applicant (even though this means younger applicants get used in two pairs). We did this because otherwise we would have to use a more stringent definition of favoring older applicants, with both older applicants but not the younger applicant getting callbacks. In this analysis, we aggregate the comparisons of older workers to younger workers and ignore the variation in whether employers prefer middle-aged and older workers to younger workers.⁵¹

Figure 7 presents the results.⁵² In general, while we find some evidence linking job-ad language to measured discrimination against younger applicants (or in favor of older ones), the results are less clearly consistent with the predictions from the industrial psychology literature. For women, we find that all of the significant associations are for personality stereotypes (similar to fig. 6). The evidence points to more discrimination against younger workers when job ads use stereotyped language associated with adaptability and less discrimination against younger workers when the job-ad language reflects the careful and dependable stereotypes. These associations point in the opposite direction to what we would have predicted on the basis of the industrial psychology literature.

For men, we find significant associations only for janitor positions. These results are less at odds with what is predicted by the industrial psychology literature. The one exception is for job-ad language related to stereotypes about hearing, for which we find evidence of more discrimination against younger workers.⁵³ For stereotyped language related to skills, we find significant effects that align with the predictions of the industrial psychology

⁵⁰ We discuss the most significant supplemental analyses here. Others are discussed in app. B.

⁵¹ Results looking at younger vs. middle-aged and younger vs. older were qualitatively similar, and results are available on request.

⁵² Table A2 reports the estimated marginal effects shown in fig. 7.

⁵³ Consistent with the caveat raised earlier, one possibility is that the language that the machine learning identifies as associated with “hearing” is also strongly associated with “listening,” which could be related to more positive stereotypes about older workers (such as careful or dependable).

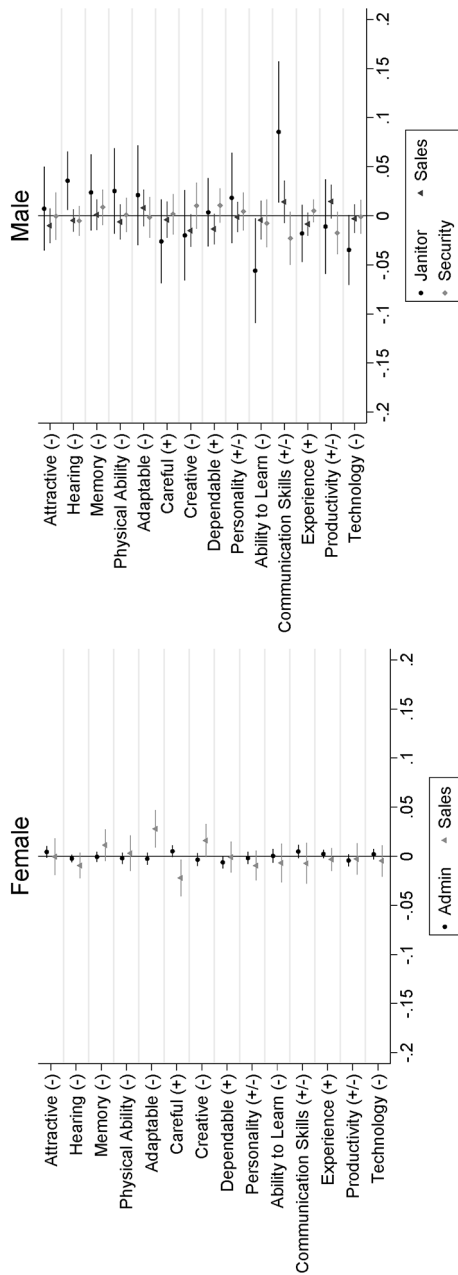


FIG. 7.—Analysis of discrimination against younger applicants. See the figure 6 legend. The difference is that the outcome is now defined as discrimination against younger applicants. Predicted signs are reported in parentheses. Positive predictions indicate that we expect to see more discrimination against younger applicants, and negative predictions indicate that we expect to see less discrimination against younger applicants.

literature. Ads that feature language more related to the ability to learn are associated with less discrimination against younger workers, and ads with language related to communication skills are associated with more discrimination against younger workers. The latter results are consistent with the positive stereotype of older workers' communication skills dominating the negative stereotype (see the conflicting stereotypes in table 3).

The results using this alternative definition of discrimination do not replicate (with the opposite sign) the results for discrimination against older workers. The simplest explanation may be that there is little discrimination in favor of older workers and little use of age-stereotyped language by employers more likely to hire older workers.

B. Alternative Corpora and Methods

Finally, we compare results using alternative methods for calculating semantic similarity. Our first method uses a naive string-matching algorithm that searches for ads containing the stereotyped words. We create 14 dummy variables corresponding to each stereotype (D_{js}). We classify each ad as containing a stereotype if the stereotyped words from tables 1–3 appear. If the stereotyped word or phrase appears in the ad, the dummy variable will take on a value of 1 and otherwise be 0. For example, our dummy variable takes on a value of 1 if the words “physically able” appear anywhere in the ad. If the words “physically able” do not appear in the ad, then the dummy variable takes on a value of 0.⁵⁴ Using these string-matching-derived dummy variables, we estimate the following equation:

$$\Pr[d_{\text{exp}}^{ij} = 1] = \alpha + \sum_s \beta_s D_{js} + X_{ij} \delta + \varepsilon_{ij}. \quad (11)$$

In this case, β_s is the change in the probability of discrimination when the job ad features the stereotyped word or phrase. The controls remain the same as in equation (10).

The second method is more closely tied to our core methods. However, we use an alternative to the CS score derived from the complete Wikipedia corpus by instead taking a narrower set of Wikipedia articles to construct the vector space. A potential concern with using the complete Wikipedia corpus is that we may sometimes measure semantic similarity from contexts far removed from age stereotypes, labor markets, and discrimination. We thus instead use an algorithm that starts from a subset of Wikipedia articles. We set this “scrapy-spider” algorithm (Kouzis-Loudas 2016) to begin with Wikipedia pages for the following: ageism, ageing, discrimination, workforce, job, employment, stereotype, sexism, recruitment, skill (labor), and human capital. The scrapy-spider algorithm then adds to our corpus the text of every page

⁵⁴ We use most of the unique words (excluding adverbs) from the “Phrasing” column in tables 1–3.

linked in these initial pages. It then repeats this process, adding the text of every page linked in those pages, which collects a total of 65,532 pages. In essence, the scrapy-spider algorithm reweights the CS scores to ignore all connections on pages more than two links away from the starting pages.⁵⁵ For example, while using the complete Wikipedia model “hearing” may be highly related to court cases, under the scrapy-spider algorithm these links are downweighted because they rarely occur within close proximity to the Wikipedia pages related to age and employment.⁵⁶

Figures 8 and 9 highlight how our estimates change when we use the string-matching (estimating eq. [11]) and the scrapy-spider (estimating eq. [10] with the alternative CS scores) algorithm, compared with our baseline estimates from the complete Wikipedia model. The confidence intervals are much larger for the string matching. Surely this is driven in part by many of the words or phrases infrequently appearing in the job ads.⁵⁷

When we compare the results from the scrapy-spider corpus to those from the complete Wikipedia corpus, we find minimal differences. The confidence intervals for our estimates always overlap, and the point estimates are often very close. Where the estimates differ, it may be for stereotypes where the semantic similarity was being driven by correlations not related to humans or workers—for example, computer memory versus human memory and court hearing versus human hearing. Consistent with this, in the different panels in figures 8 and 9, the confidence intervals overlap less closely for cases including memory, technology, communication, and hearing. Moreover, there are often cases for which estimates shift toward more discrimination against older workers when ads use language highly related to the stereotype; examples include technology for women in sales (middle-aged and older), adaptable for older men in security, and technology for older men in sales. However, the estimates do not uniformly shift toward more evidence of age discrimination

⁵⁵ We chose to set the algorithm to use two links because the number of links grows exponentially.

⁵⁶ Table A3 compares the words nearest the 95th percentile in an ad using this scrapy-spider algorithm and using the complete Wikipedia corpus, for selected stereotypes. The full table is available on request. We see that the words at the 95th percentile seem more related to the employment context using the scrapy-spider algorithm (including for the “hearing” example just noted) but that the differences can be subtle. We also show the words used in the string matching. Comparing these two algorithms to the string matching highlights the value of measuring relatedness and not simply the presence of specific words, since all of the words shown from the Wikipedia-derived models that do not match exactly would be coded as zero in eq. (11).

⁵⁷ For example, the words “careful” and “memory” appear in only 35 and 31 job ads, respectively. More generally, for many gender-age-occupation cells there are cases where no ad featured the words on which we string match. This is most common for the words “memory,” “careful,” and “creative.”

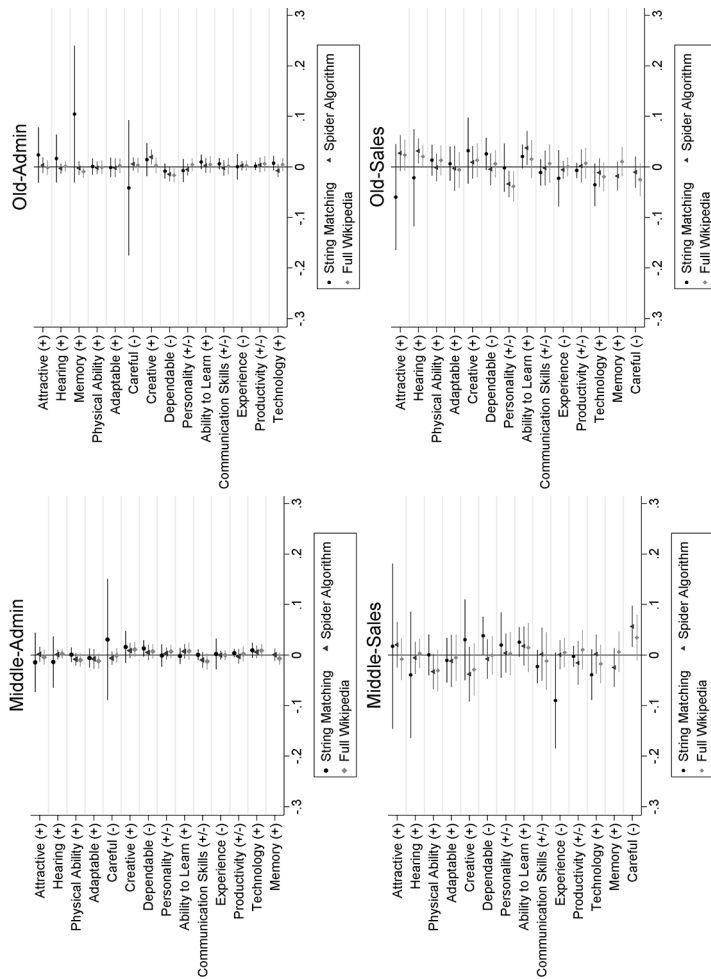


FIG. 8.—Comparing results with alternative corpora/methods, women. This figure reports the estimated marginal effects and 95% confidence intervals as we vary the corpora or method used to identify stereotyped language. The string matching reports the marginal effect of the dummy variable indicating that the stereotype appeared in a job ad. Positive predictions indicate that we expect to see more discrimination against older applicants, and negative predictions indicate that we expect to see less discrimination against older applicants. For age-occupation-gender cells where the word does not appear in any ad or is perfectly correlated with discrimination, no coefficient is reported. This occurs most often for memory, careful, and creative. The spider algorithm model and the complete Wikipedia models report the estimates from equation (10), with the complete Wikipedia model being identical to the baseline results. The string matching uses equation (11). Standard errors are clustered at the job-ad level.

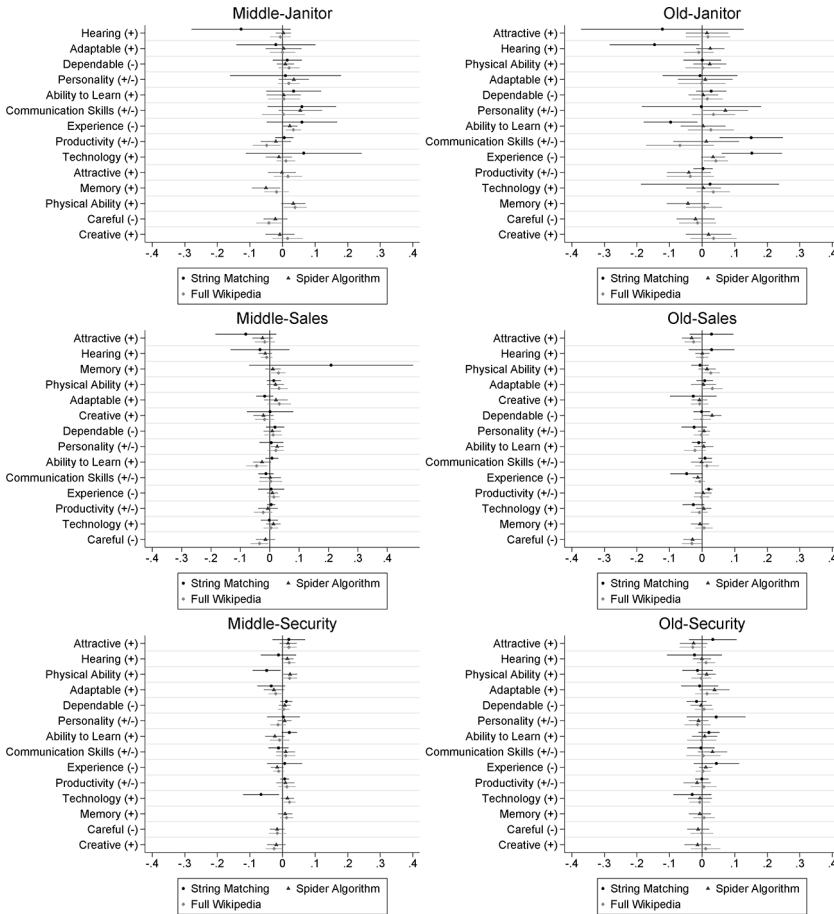


FIG. 9.—Comparing results with alternative corpora/methods, men. See the figure 8 legend.

when we classify job-ad language using the narrower corpus from the scrapy-spider algorithm.⁵⁸

VIII. Discussion and Conclusions

We develop new machine learning techniques for analyzing complex textual data, which we apply to job ads collected in a large-scale resume-correspondence study of age discrimination. We combine the machine learning analysis of the text of job ads with experimental measures of age discrimination from the correspondence study to examine whether phrases

⁵⁸ Although one advantage of using complete Wikipedia is that it avoids data mining, this could be avoided using scrapy spider by prespecifying the pages to be used if an experiment were planned that would download job ads anew.

in the job ads that are strongly related to ageist stereotypes predict age discrimination in hiring.

The evidence suggests that ageist stereotypes in job ads are related to employers' decisions not to call back older applicants. For both men and women and across different occupations, we find evidence that employers who do not call back older applicants but do call back younger applicants (or vice versa) use phrases in their job ads that are related to ageist stereotypes. For men we find evidence that age stereotypes about all three categories we consider—health, personality, and skill—predict age discrimination, and for women age stereotypes about personality predict age-related differences in hiring. In general, the evidence is much stronger for men, and our results for men are quite consistent with the industrial psychology literature on age stereotypes, with many negative age stereotypes reflected in job-ad language predicting more hiring discrimination against older workers and some positive age stereotypes predicting the opposite.

The stronger and more robust results for men than for women suggest that stereotypes in job ads may play a more prominent role in generating observed hiring discrimination against older men than against older women, even though the correspondence study found stronger evidence of hiring discrimination against older women. A similar puzzle is that the evidence points to larger effects of stereotypes in job ads for middle-aged than for older men, despite the experimental evidence providing stronger evidence of age discrimination in hiring against older men. Why might stereotyped language matter more to older workers when measured discrimination is lower?

One potential explanation is that the motivation to use age stereotypes to discourage workers from applying may interact with the effectiveness of age discrimination laws. If age discrimination laws are less effective for a particular subgroup of older workers, then employers who do not want to hire from this subgroup do not have as great an incentive to use ageist stereotypes in job ads to shape the applicant pool to try to avoid detection of discriminatory behavior. Age discrimination laws may be less effective for older women than for older men because of the difficulty of bringing to court intersectional discrimination claims based on being both female and older (McLaughlin 2019). Age discrimination laws may also be less effective at deterring discrimination against older men than middle-aged men because the much smaller number of older job applicants relative to middle-aged job applicants implies that the same hiring rate difference relative to young workers is less likely to be statistically significant for the older applicants.⁵⁹

⁵⁹ For example, suppose a company expects to get 100 young applicants, 100 middle-aged applicants, and only 10 older applicants. (This is consistent with evidence reported in Neumark, Burn, and Button [2019] as well as with the much lower employment rate at ages near 65 than ages near 50.) If the company were to hire 20 young, 10 middle-aged, and one older applicant, the relative hiring rates would be the same for both middle-aged and older applicants relative to young applicants, but the middle-aged

In addition, the list of stereotypes we compiled from the industrial psychology literature may drive the differences by gender, for two reasons. First, the industrial psychology literature on age stereotypes primarily identifies stereotypes associated with men, in which case the stereotypes may be less salient for women (or different stereotypes than those we study may matter more). Second, the stereotypes relevant in more traditionally female jobs (like administrative assistants) may be more challenging to express in job ads, leading to weaker relationships between the underlying stereotypes and phrases from the job ads.

Our evidence that measured age discrimination is related to ageist stereotypes in job ads has potential policy implications. If older workers know the relationship between ageist job-ad language and hiring discrimination and apply less to stereotyped job ads, then comparing hiring rates and application rates by age can understate hiring-related age discrimination. On the other hand, our evidence has potentially constructive implications for enforcing age discrimination laws. If discriminatory employers use ageist language in their job ads, then barring such language may increase applications from older workers, making it harder for hiring discrimination to go undetected. And testing for age stereotypes in job ads could be used to detect firms that may discriminate on the basis of age in hiring decisions. Of course, the methods we develop also could be applied to discrimination against other groups.

One limitation of our work is that we study only age stereotypes that appear in job ads. Employers could have other stereotypes about older workers that affect hiring but on which our evidence is silent. On the other hand, thinking back to our two primary hypotheses—that employers who discriminate on the basis of age use stereotyped language to shape the applicant pool and that employers statistically discriminate on the basis of stereotypes about older workers' ability to meet job requirements—we may be most interested in the stereotypes expressed in job ads. For example, if age-related stereotypes in job ads signal the dimensions along which employers statistically discriminate in hiring, then these are the stereotypes that need to be assessed against the RFOA criterion. Moreover, if some stereotypes are identified in the lab but not expressed in real-world job ads, they may simply not be very relevant to real-world labor market decisions.

A second limitation is that we do not study the behavior of job applicants in response to ageist stereotypes in job ads. As we have argued, these responses can be important, reducing the need for a discriminatory employer to hire older applicants at a lower rate. As we have shown, this can create a

vs. young difference would be more strongly significant because of the greater number of observations. The intersectionality issue for older women vs. older men can also be interpreted in terms of statistical evidence because the inability to present evidence based on both age and sex can preclude evidence of a negative interaction between being older and female.

bias against finding evidence that stereotyped job-ad language predicts age discrimination in hiring, which could, in principle at least, explain why in many cases we do not find such evidence.

A key contribution of our techniques is that they can be adapted to other contexts. The most direct application, perhaps, is to future audit or correspondence studies of labor market discrimination. There has been an explosion in these kinds of studies in recent years,⁶⁰ but one could reasonably argue (see Neumark 2018) that it is time for these studies to move beyond documenting differences in outcomes and learning more about the underlying behavior. In nearly all of these studies, some type of job-ad postings are available as objects of study. It may also be possible to apply our methods to studies of discrimination in other markets—such as housing or health care—depending on what kind of information is included in the ads or postings used in the market. With relatively few changes to our methods, researchers could test for relationships between the usage of stereotyped language and the discrimination these studies measure. Moreover, these language-processing techniques could be useful in studying discrimination in different parts of the process of hiring or other employment decisions, such as recommendation letters or employee evaluations.⁶¹ Researchers have also applied methods related to ours to detect bias in language in other settings, such as judges' decisions (Ash, Chen, and Ornaghi 2020).

Moreover, related methods could be used in many other areas of labor economics. For example, a recent study presents a creative analysis of text from job advertisements and course syllabi to study how education responds to skill demands (Börner et al. 2018). Anastopoulos et al. (2019) use machine learning techniques to study the effects of immigration on demands for different kinds of jobs. And research we cited earlier uses the text of job postings from sources like CareerBuilder.com and Burning Glass Technologies to test different hypotheses about the labor market (e.g., on job search, Banfi and Villena-Roldán 2019; Marinescu and Wolthoff 2020). In our view, the number of potential applications in labor economics, especially combining job postings with other types of text, is enormous.

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⁶⁰ See the recent book by Gaddis (2018) and the register of studies since 2005 maintained by Baert at https://users.ugent.be/~sbaert/research_register.htm.

⁶¹ For a discussion of research on letters of recommendation, see Madera, Hebl, and Martin (2009).

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